

Prediction of Chronic Disease Risk Using Nutrition, Lifestyle & Clinical Data from The Irish Longitudinal Study on Ageing (TILDA)



Abderahman Haouit

CSIS Research Centre

Department of Computer Science and Information Systems

Faculty of Science and Engineering

University of Limerick

Submitted to the University of Limerick for the degree of

Artificial Intelligence (MSc) 2026

-
1. Supervisor: Dr. Alison O'Connor
Computer Science & Information Systems
University *of* Limerick
Ireland

Abstract

Background: Chronic diseases such as Cardiovascular Disease (CVD), Diabetes Mellitus (DM), Frailty (FR), and obesity represent major contributors to morbidity in older adults. While traditional prediction models focus heavily on clinical and functional indicators, the contribution of nutritional and lifestyle factors to early risk detection remains underexplored.

Methods: This thesis applies machine learning techniques to data from The Irish Longitudinal Study on Ageing (TILDA), a national representative longitudinal study of older adults. Five waves of socio-demographic, clinical, nutritional, psychological, and behavioural data, with selected variables from the externally produced Research and Development Corporation (RAND) + Harmonised TILDA dataset, together were systematically cleaned and standardised using a multi-stage preprocessing pipeline. Predictive models, including (*models x , y , z*) were developed to estimate the risk of chronic disease and FR-related outcomes. Model performance was evaluated using (*mention examples*), and feature importance was used to assess predictive value.

Results: TBD

Conclusions: TBD

Keywords: ageing, (Artificial Intelligence (AI)), Machine Learning (ML), predictive modelling, Frailty (FR), chronic disease, nutrition, The Irish Longitudinal Study on Ageing (TILDA)

Declaration

I Abderahman Haouit declare that I have produced this thesis without the prohibited assistance of third parties and without making use of aids other than those specified; notions taken over directly or indirectly from other sources have been identified as such. This thesis has not previously been presented in identical or similar form to any other Irish or foreign examination board.

The thesis work was conducted from 2025 to 2026 under the supervision of Dr. Alison O'Connor at University of Limerick.

Limerick, 2026

Acknowledgements

I would like to acknowledge my supervisor, Dr. Alison O'Connor, for her guidance, support, and for providing a well organised environment throughout the implementation of this project.

I would also like to thank the UL Science and Engineering Research Ethics Committee for approving the ethical aspects of this research. Special thanks to the Irish Social Science Data Archive (ISSDA) team for facilitating access to the TILDA dataset, collected by Trinity College Dublin.

Your dedication goes here. Open the dedication.tex file found at:
0_frontmatter/dedication.tex

Contents

List of Tables	vi
List of Figures	vii
List of Abbreviations	ix
1 Introduction	1
1.1 Research problem and motivation	2
1.2 Research aims and objectives	3
1.3 Contributions of the research	4
1.4 Thesis Outline	4
2 Literature Review	6
2.1 Methodology	6
2.1.1 Search strategy and database screening	7
2.1.2 Quality and relevance evaluation	9
2.1.3 Synthesis and Integration	10
2.2 Frailty and physical function	12
2.2.1 Probabilistic interpretation of frailty	12
2.2.2 Bayesian updating of frailty risk	13
2.3 Mental health in older adults	13
2.4 Nutrition and biological ageing	14
2.4.1 Bayesian perspective on nutritional biomarkers	15
2.4.2 Contextual and integrative considerations	15
2.5 Machine Learning in ageing and chronic disease	16
2.5.1 Feature sets in ML studies	16

CONTENTS

2.5.2	ML models used	16
2.5.3	Performance and Validation	17
2.5.4	Explainability in ML for ageing	17
2.6	Gaps in the Literature	18
2.7	Chapter Summary	19
3	Technical Approach	21
3.1	Study Design	21
3.2	Cohort Definition	21
3.3	TILDA Data Collection and Screening	22
3.4	Thematic Grouping	27
4	Implementation Details	31
4.1	Cleaning Engine	32
4.1.1	Stage 1: Variable scenario mapping and validation	33
4.1.2	Stage 2: Deterministic six step cleaning sequence	34
4.1.3	Scenario steps and their technical focus	36
4.2	Cleaning Validation	38
4.2.1	Pairwise correlation validation	38
4.2.2	Model Validation	39
5	Empirical Evaluation	41
6	Conclusion	42
6.1	Summary	42
6.2	Conclusions	42
6.2.1	Data availability Statement	42
6.2.2	Ethics Statement	42
6.3	Contributions	43
6.4	Future Work	43
A	Python code for exporting PMF metadata to CSV	44
B	Scenario Cleaning Pipeline Tables	47

CONTENTS

C First Stage: Scenario Variable Mapping	50
D Second Stage: Deterministic six step cleaning engine	55
References	60

List of Tables

2.1	Database searches and number of studies per database 2.3 [1]. . .	8
2.2	Quality evaluation of Systematic Literature Review (SLR)s [2]. . .	10
2.3	Summary of selected studies on ageing using TILDA.	11
3.1	Dimensions of raw TILDA and harmonised datasets (prior to filtering).	24
3.2	Pseudonymised Microdata File (PMF) Derived variable prefixes [3].	28
3.3	Thematic grouping of longitudinal features.	30
4.1	Input shape for cleaning pipeline (post manual variable filtering).	31
4.2	Inconsistent category ordering for the same feature <i>ph009</i> across waves and its impact on positional (order-based) label encoding. .	37
4.3	Directional validation of the seven features explaining 95% of Gait Speed Reserve (GSR) variance as identified by [4].	40
B.1	Initial Raw Feature Counts Categorised by Nine Scenarios (Pre Cleaning)	47
B.2	Scenario Count Progression: Post Step 1 (Empty / Blank Handling)	48
B.3	Scenario Count Progression: Post Step 2 (Sentinel Standardisation)	48
B.4	Scenario Count Progression: Post Step 3 (Numeric Cleaning) . . .	48
B.5	Scenario Count Progression: Post Step 4 (Censored / Ordinal Mapping)	49
B.6	Scenario Count Progression: Post Step 5 (Binary Classification) .	49
B.7	Final Dfs Cleaned: Post Step 6 (Multi-Class / Mixed Parsing) . .	49

List of Figures

2.1	Publications per year.	7
2.2	PRISMA chart for Systematic Review (SR)s showing the study selection process for the systematic mapping and integrative review [5].	9
3.1	Flowchart of the five step workflow for variable screening and filtering. Placeholder counts indicate the progression of remaining variables per dataset from raw to filtered datasets after each step, ready for automated preprocessing.	26
4.1	Flowchart of the final six step cleaning pipeline applied to all TILDA waves. See full transformation tables in Appendix B.7 and D for detailed scenario progression code at each step.	35
4.2	Correlation heatmap for top 7 predictors (Wave 3).	39

List of Abbreviations

AI	Artificial Intelligence
AUC	Area Under the ROC Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
BMI	Body Mass Index
BP	Blood Pressure from Sphygmomanometer
brainPAD	Brain predicted age difference
CAPI	Computer Assisted Personal Interview
CES-D	Center for Epidemiologic Studies Depression Scale
COG	Scales and Questions on Cognition
CRT	Choice Reaction Time (Cognitive Test)
CV	Cross Validation
CVD	Cardiovascular Disease
DIS	Disability, Functional Impairment and Helpers
DM	Diabetes Mellitus
DNN	Deep Neural Network
EU	European Union
F1	F1-score (harmonic mean of precision and recall)
FOF	Fear of Falling
FR	Frailty
GBM	Gradient Boosting Machine
GDPR	General Data Protection Regulation
HADS	Hospital Anxiety and Depression Scale
HGS	Hand Grip Strength

List of Abbreviations

IADL	Instrumental Activities of Daily Living
INC	Income Variables
IPAQ	International Physical Activity Questionnaire
ISSDA	Irish Social Science Data Archive
JSON	JavaScript Object Notation
MD	Medication Data
MGS	Maximum Gait Speed
MH	Scales and Questions on Mental Health and Wellbeing
ML	Machine Learning
MVPA	Moderate-to-Vigorous Physical Activity
NOS	Newcastle-Ottawa Scale
OR	Odds Ratio
PA	Physical Activity
PMF	Pseudonymised Microdata File
RAND	Research and Development Corporation
Recall	Recall (true positive rate)
RF	Random Forest
SCQ	Self Completion Questionnaire
SCR	Screening Tests
SHAP	Shapley Additive Explanations
SLR	Systematic Literature Review
SOC	Social Variables
SPSS	Statistical Package for the Social Sciences
SR	Systematic Review
TILDA	The Irish Longitudinal Study on Ageing
TUG	Timed Up and Go (mobility test)
UGS	Usual Gait Speed
WHO	World Health Organization

List of Abbreviations

Chapter 1

Introduction

Ageing populations experience substantial changes across physical, nutritional, cognitive, and psychosocial domains, each shaping vulnerability to chronic disease and functional decline [6–8]. Understanding how these multidomain factors interact over time is essential for early risk identification.

The financial and societal burden is substantial. In Europe, poor diet is a major risk factor for non communicable diseases, contributing billions in health-care costs annually [9]. In Ireland, malnutrition alone is estimated to cost over €1.4 billion per year [10]. In response to these challenges, international public health frameworks, including the World Health Organization (WHO) *Global Action Plan* on noncommunicable diseases and Ireland’s *Healthy Ireland* framework, highlight nutrition and lifestyle factors as central components of disease prevention strategies.

European regulatory frameworks such as the General Data Protection Regulation (GDPR) and proposed European Union (EU) Artificial Intelligence (AI) Act emphasise transparency and highlight the need for interpretable Machine Learning (ML) models in healthcare.

Traditional risk prediction models have largely relied on clinical biomarkers such as Blood Pressure from Sphygmomanometer (BP) and lipid profiles. Yet, growing evidence shows that broader predictors, including nutrition [11], Physical Activity (PA) [12], Scales and Questions on Mental Health and Wellbeing (MH) [13], and Social Variables (SOC) [14] improve predictive performance. Es-

1. INTRODUCTION

timating the likelihood of developing a condition given certain risk factors helps quantify the influence of these predictors in clinical decision making.

TILDA¹ provides a national representative cohort of adults aged 50+ in Ireland, surveyed repeatedly across multiple domains. For cross study comparability and methodological consistency in model development, selected variables from the externally produced RAND Harmonised TILDA dataset (available for Waves 1 and 2) were included.

The harmonised dataset provides standardised variable definitions and documented construction rules, which are particularly relevant in a machine learning setting where models are trained and evaluated on partitioned data. Its inclusion supports robust feature engineering, reproducible train validation splits, and alignment with prior ageing related ML studies that adopt harmonised frameworks. The specific role of harmonised variables within the modelling pipeline is detailed in Chapter 3.

Despite these advantages, existing ML research on chronic disease prediction faces persistent limitations: over reliance on clinical predictors, limited incorporation of lifestyle and SOC variables, predominantly cross sectional study designs that fail to capture trajectories, and a lack of interpretability consistent with emerging AI governance standards.

This thesis addresses these gaps by leveraging longitudinal TILDA data across multiple domains, using interpretable ML methods to combine modifiable health behaviours, psychosocial factors, and established clinical measures to enhance prediction of ageing related outcomes.

1.1 Research problem and motivation

Despite increasing research and clinical interest in lifestyle, nutrition factors and advances in ML methods, approaches to chronic disease prediction in older adults remain limited in several ways.

¹TILDA's dataset are pseudonymised and accessed under license via ISSDA. Use adheres to UL S&E ethics approval and data retention requirements.

- **Limited Integration of Multidomain Predictors:** Existing models focus primarily on clinical measurements and often neglect modifiable behavioural, dietary, and psychosocial factors [11, 15].
- **Underuse of Longitudinal Dynamics:** Few studies leverage repeated measures to capture developments of risk, such as FR progression [16]. Probabilistic reasoning can help estimate the likelihood of developing a condition at a future time given earlier measurements, for example, calculating the probability $P(\text{FR at wave 5} \mid \text{UGS at wave 1, Nutrition at wave 2})$.

This approach helps quantify how early indicators influence later outcomes and supports better feature selection and timing of interventions in predictive models.

- **Lack of Explainability:** Many ML models are not interpretable, limiting clinical adoption and failing to meet emerging regulatory requirements for trustworthiness AI in healthcare [17].

Addressing these challenges is essential to improve predictive accuracy, support evidence based interventions, and ensure alignment with evolving policy and regulatory priorities.

1.2 Research aims and objectives

The goal of this thesis is to develop and validate ML models that integrate nutrition, lifestyle, and MH features with clinical data to improve prediction of chronic disease and ageing related outcomes in a representative cohort of older adults. This goal will be achieved through the following objectives:

- Review existing applications in ML for chronic disease prediction in ageing populations, with emphasis on gaps in lifestyle and nutrition integration.
- Develop a reproducible pipeline to preprocess and align TILDA waves 1–5, and engineer multidomain features (nutrition, activity, mobility, MH, clinical indicators).

1. INTRODUCTION

- Develop and compare baseline statistical models (e.g., logistic regression) with advanced interpretable ML methods such as Random Forest (RF) and Gradient Boosting Machine (GBM).
- Evaluate models using predictive classification performance metrics including Area Under the ROC Curve (AUC), calibration and error measures (accuracy, precision, Recall (true positive rate) (Recall), F1-score (harmonic mean of precision and recall) (F1)), perform ablation analyses to assess the incremental value of lifestyle and nutrition predictors.
- Identify actionable predictors (e.g., vitamin D status, UGS reserve) and highlight modifiable risk factors for intervention.
- Document ethical, governance, and reproducibility practices for handling longitudinal pseudonymised health data.

1.3 Contributions of the research

This research makes contributions at three levels:

- **Empirical:** Demonstrates how behavioural, dietary, and other MH features complement established clinical measures to enhance predictive performance.
- **Methodological:** Provides a reproducible pipeline for aligning longitudinal TILDA data across domains and integrating diverse features into ML workflows.
- **Clinical:** Highlights actionable modifiable factors alongside traditional clinical indicators for targeted interventions and policy.

1.4 Thesis Outline

In *Chapter 2* a literature review on ML prediction of chronic disease in ageing populations, with emphasis on lifestyle, nutrition predictors and methodological

considerations, is presented. *Chapter 3* discusses the TILDA study design, data preprocessing, alignment, and feature engineering. In *Chapter 4* the modelling framework is outlined. This includes baseline and advanced ML, hyperparameter tuning, validation, and evaluation metrics. *Chapter 5* shows the model result, including model comparisons, assessments of feature importance, and an ablation analysis. In *Chapter 6* the implications, limitations, and ethical considerations are discussed, and future directions for the research are proposed. Additional supporting documents are included in appendix A.

Chapter 2

Literature Review

This chapter reviews literature on ageing, chronic disease, and predictive modelling using TILDA (introduced in Chapter 1). It focuses on three interrelated domains central to later life health: FR, physical function, MH, and nutrition within the context of biological ageing. This review focuses on FR, MH, and nutrition as key determinants of later life outcomes. In line with the WHO healthy ageing framework, research using TILDA highlights FR, MH, and nutrition as interconnected determinants of later life outcomes. The following subsections examine each domain in turn.

2.1 Methodology

Several sources describe how systematic and integrative reviews are commonly carried out in fields such as software engineering, health, and education. The approach followed key principles for systematic mapping, for example, defining research questions, identifying relevant studies through structured database searches, applying inclusion and exclusion criteria, and categorising results as outlined in [1]. The integrative literature review [18] helped structure and connect findings across studies, while [19, 20] guided the organisation of ideas. In related methodological work, [2] clarified how SRs can identify research gaps, and [21] showed how mapping and synthesis highlight trends and future directions. This combined guidance informed a structured yet flexible methodology suitable for ageing in AI, and health research.

2.1.1 Search strategy and database screening

The main topic of interest was chronic disease prediction using nutrition, lifestyle, and clinical data from TILDA. Relevant databases (e.g., ScienceDirect, PubMed) were searched using combinations of keywords such as “Irish Longitudinal Study on Ageing”, “machine learning,” “chronic disease”, “nutrition”, “diet”, and “physical activity” (see table 2.1).

The search covered studies published from 2011 to 2025. Figure 2.1 summarises the number of studies identified per year, highlighting trends and gaps over time.

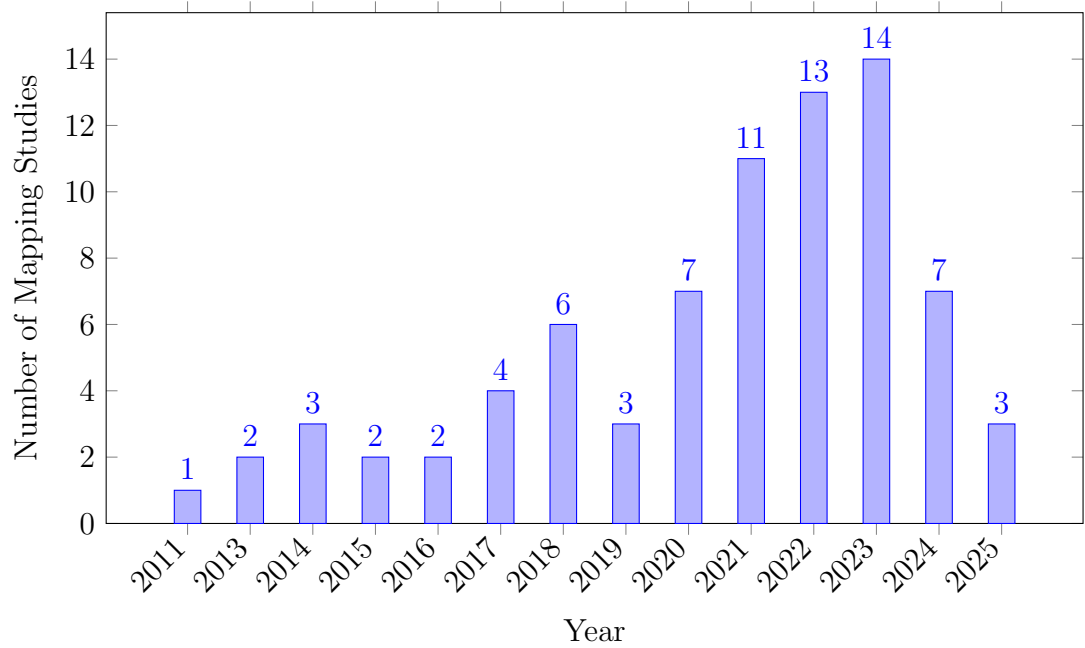


Figure 2.1: Publications per year.

Some search strings returned few or irrelevant results, particularly for ML related TILDA studies, highlighting a key research gap. Iterations continued until suitable combinations consistently returned relevant studies. Duplicates were removed using Zotero’s reference management feature. The remaining studies were individually and manually checked for relevance to “ageing,” “chronic disease,” and ML. Using Zotero’s notes functionality, the initially created table 2.3 captured key elements for each paper, including model type, features used, and

2. LITERATURE REVIEW

Table 2.1: Database searches and number of studies per database 2.3 [1].

Database	Search	Results
Oxford Academic	(TILDA OR Irish Longitudinal Study on Ageing)AND (chronic disease OR diabetes OR cardiovascular) AND (nu- trition OR diet OR “physical activity”)	1081
	(TILDA OR “Irish Longitudinal Study on Ageing”) AND (“machine learning” OR “predictive model” OR “risk pre- diction”)	71
IEEE Xplore	TILDA OR “Irish Longitudinal Study on Ageing”	56
	(TILDA OR ”Irish Longitudinal Study on Ageing”) AND (“machine learning” OR “predictive model” OR “risk pre- diction”)	7
ScienceDirect	(TILDA OR “Irish Longitudinal Study on Ageing”) AND (“chronic disease” OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR “physical activity”)	257
	(TILDA OR “Irish Longitudinal Study on Ageing”) AND (“machine learning” OR “predictive model”) AND (“chronic disease” OR diabetes) AND (nutrition OR “physical activ- ity”)	20
PubMed	(TILDA OR “Irish Longitudinal Study on Ageing”) AND (“chronic disease” OR cardiovascular OR diabetes) AND (nutrition OR diet OR lifestyle OR “physical activity”)	27
	(TILDA OR “Irish Longitudinal Study on Ageing”) AND (“machine learning” OR “predictive model” OR “risk pre- diction”)	9

main findings for each paper in the collection, as illustrated in the PRISMA flow diagram (Figure 2.2).

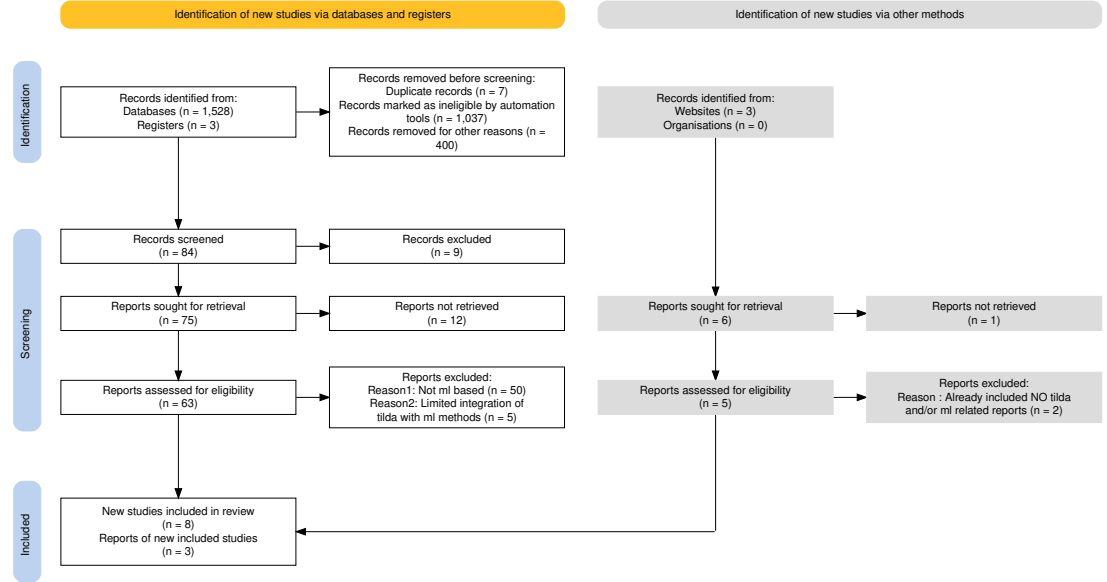


Figure 2.2: PRISMA chart for SRs showing the study selection process for the systematic mapping and integrative review [5].

2.1.2 Quality and relevance evaluation

To systematically assess the quality of the included studies, a customisable version Newcastle-Ottawa Scale (NOS) was applied; see table 2.2. The NOS evaluation studies across three domains: *Selection* (S1–S4), *Comparability* (C1–C2), and *Outcome* (O1–O3). *Selection* criteria assess the representativeness of the sample, clarity of exposure measurement, and temporal alignment of outcomes. *Comparability* examines the control of key confounders (variables that may influence both the predictor and outcome), while *outcome* evaluates measurement validity, follow up adequacy, and completeness. Each domain is scored 0–1, recall the Bernoulli random variable (0 = criterion not met, 1 = criterion met), and the total score reflects the overall study quality.

This initial table guided a further refinement of the results into two sub tables: (1) studies critically related to TILDA, (2) studies using TILDA data ML for

2. LITERATURE REVIEW

predictive contexts, which revealed minimal to no integration with TILDA.

Table 2.2: Quality evaluation of SLRs [2].

Reference	S1	S2	S3	S4	C1	C2	O1	O2	O3	Total
(Xu, 2023 [22])	1	1	1	1	1	1	1	1	1	9
(Donoghue, 2023 [17])	1	1	1	1	1	1	1	0	1	8
(Moguilner, 2021 [16])	1	1	1	1	1	0	1	1	1	8
(Davis, 2021 [6])	1	1	1	1	1	0	1	1	0	7
(Boyle, 2021 [23])	1	1	1	1	1	0	1	0	1	7
(Tzemah-Shahar, 2022 [24])	1	1	1	0	1	0	1	1	1	7
(A. Shirsath, 2024 [25])	1	1	1	1	1	1	1	0	1	8
(Romero-Ortuno, 2021 [26])	1	1	1	1	1	1	1	1	1	9
(Bardon, 2020 [27])	1	1	1	0	1	0	1	1	1	7
(Zhang, 2025 [28])	1	1	1	1	1	1	1	0	1	8
(Sutin, 2023 [29])	1	1	1	1	0	0	1	1	1	7

Table 2.3 summarises selected studies on ageing using TILDA and ML, reflecting key gaps and opportunities identified during systematic mapping. These steps were aligned with the research matrix created at the beginning, ensuring that the refinement of studies and synthesis of findings stayed closely tied to the original research goals.

2.1.3 Synthesis and Integration

Using the NOS scores, each study was annotated in Zotero with details on dataset, key variables, analytical approach, and main findings, facilitating a consistent and systematic evaluation across studies. For the systematic component, data was systematically extracted and synthesised to identify trends (e.g., Markov models, logistic regression, 5-fold Cross Validation (CV)) and research gaps [2], complemented by DoctoralWriting resources [30, 31], which clarified how to define research questions and develop logical arguments.

The integration of a research matrix with systematic mapping and integrative synthesis ensured that the methodology was well suited to capturing the complexity of ageing, nutrition, chronic disease risk and predictive modelling studies focused primarily on TILDA based studies, but due to the limited number of ML

2.1 Methodology

Table 2.3: Summary of selected studies on ageing using TILDA.

Theme	Author/year	Dataset	Variables/features	Methods/model type	Key Findings/outcome	Limitations
FR	Romero-Ortuno (2021) [26]	Wave 1–5	FR state, FR phenotype (FP) components (Exhaustion(EX), Unexplained weight loss(WL), Weakness(WK), Slowness(SL), Low physical activity(LA))	Multi state Markov model, longitudinal 8 year follow up	32% of pre frail to non frail; FR strongly predicted 31% mortality	Self report bias; observational design
Falls + ML	Donoghue (2023) [17]	Wave 1–3	Gait, mobility, medications, history of falls	Conditional inference forest; 5-fold CV; undersampling for imbalance	Accuracy up to 69.7%; for adults aged 50–64.	Area Under the Receiver Operating Characteristic Curve (AUROC) values were ≤ 0.69 and PRAUC ≤ 0.39 indicating low predictive accuracy; no external validation; cross sectional; Class imbalance
Fear of Falling (FOF)	Zarudsky (2014) [32]	Wave 1	Age, sex, fall history	Cross sectional prevalence analysis	23.3% prevalence; higher in women; 70% with FOF had no recent falls in the last 12 months	Self reported; cross sectional
Functional Mobility	Huang (2019) [33]	Wave 1–2	Measured by Timed Up and Go (mobility test) (TUG), Hand Grip Strength (HGS), physical activity, vision, Body Mass Index (BMI), age	Linear regression, stratified analyses	Age modifies effects; PA and grip protect mobility; vision benefit only in middle aged (Beta = -0.032, P = 0.002)	Observational; 2 year follow up
Depression	Jacob (2023) [34]	Wave 1–2	Falls, FOF, HADS-A, Center for Epidemiologic Studies Depression Scale (CES-D)	Prospective logistic regression	Falls predicted incident anxiety Odds Ratio (OR)=1.58 and depression OR=1.43	Self reported falls; short follow up; mediation effect partially explored
	Laird (2023) [15]	Wave 1–5	Moderate-to-Vigorous Physical Activity (MVPA), walking, sociodemographics	Longitudinal regression; mixed model	Higher physical activity reduced depressive symptoms; dose response effect	Self reported PA; observational; limited causal inference
Nutrition	Murphy (2023) [11]	Wave 1–5	Plasma lutein, zeaxanthin	Linear/ordinal logistic regression	Higher carotenoids predicted slower FR progression Lower odds of becoming frail (ORs = 0.57–0.89)	Observational; not predictive of longitudinal muscle changes
	Bardon (2020) [27]	Wave 1–2	Hospitalisation, functional status, SOC support, sex, falls	Logistic regression; sex stratified analyses	Functional decline, hospitalisation, poor support increased malnutrition risk; sex differences	Observational(short follow up (2 years)); limited generalisability ;65 years, self reported
ML	Xu (2023) [22]	Wave 1–3	105 predictors from 40,000+ variables (SOC, clinical, mental, family)	Stacked ensemble: RF, XRT, GLM, GBM, Deep Neural Network (DNN); Shapley Additive Explanations (SHAP) explainability	AUC=0.854 for 2 year DM prediction; top predictors LDL, cirrhosis, sex	No external validation; short follow up; ensemble complexity limits deployment
	Davis (2021) [6]	Wave 3	Age, HGS, chair stands, BMI, cognitive tests	Histogram gradient boosting regression; SHAP	r^2 test: UGS $\approx$ 0.41, Maximum Gait Speed (MGS) $\approx$ 0.46, GSR $\approx$ 0.21	Modest predictive power; cross sectional; no external validation
	Zhang (2025) [28]	Longitudinal IMAGEN cohort	Psychopathology, personality, cognition, substance use, environment	Regularised logistic regression; diagnostic	Longitudinal AUC: ED=0.71, MDD=0.64, AUD=0.67; shared transdiagnostic predictors	Limited age range; no external replication
	Boyle (2021) [23]	Wave 3	Brain predicted age difference (brainPAD); processing speed; attention; cognitive flexibility; verbal fluency	Linear regression, correlation analyses	Higher brainPAD associated with lower cognitive performance across multiple domains	Cross sectional; no causal inference; limited sample

2. LITERATURE REVIEW

applications within TILDA, additional studies on older adult populations and chronic disease were included.

2.2 Frailty and physical function

FR reflects declining physiological reserve, providing a framework for understanding physical vulnerability in ageing. In TILDA, it is captured through both the FR phenotype and the FR index, with UGS, HGS, and mobility decline serving as practical predictors. These measures strongly predict hospitalisation and mortality.

2.2.1 Probabilistic interpretation of frailty

One can argue that from a probabilistic perspective, the chance of an adverse outcome (such as mortality, Y) given observed features (such as FR markers, X) is a conditional probability $P(Y | X)$. This allows us to deduce which FR features are most predictive of adverse outcomes. Features may be independent, dependent, or mutually exclusive.

The presence or absence of metabolic syndrome represents a mutually exclusive event that can inform risk models.

$$P(X_i \cap X_j) = P(X_i)P(X_j)$$

for independent features X_i and X_j , guiding feature selection and reducing redundancy in model inputs. This reflects the likelihood that an individual with certain FR characteristics will experience a particular outcome.

Evidence suggests that FR is dynamic, not irreversible. For example, multi state modelling showed that many prefrail adults transition back to non frail, although established FR strongly predicts mortality [26]. Metabolic syndrome accelerates the development of FR, suggesting a metabolic pathway of vulnerability [35]. The presence of metabolic syndrome modifies the conditional probability $P(\text{FR} | \text{Metabolic Syndrome})$ and may increase the likelihood of adverse outcomes over time. Another detail is that age matters: obesity predicts mobility decline earlier in life, whereas later in life, traits that protect against this, including HGS, become more significant [33].

These findings partly conflict: FR appears both reversible and progressive, depending on risk context. This complexity highlights the need for probabilistic models that can update beliefs about risk as new features are observed, as formalised by Bayes theorem:

$$P(Y | X) = \frac{P(X | Y)P(Y)}{P(X)}$$

where $P(Y | X)$ is the posterior probability of outcome Y given features X .

2.2.2 Bayesian updating of frailty risk

Causal inference is limited by observational designs, and predictors are frequently examined separately, ignoring interactions. Most FR models have only been with the TILDA dataset, and with this study that focuses on the same TILDA but uses both longitudinal TILDA waves and harmonised variables to improve cross study comparability, enabling for a more robust training validation splits in ML models, and explores richer predictive patterns over time. These gaps point to the need for multidimensional approaches. Traditional regression models may not fully capture age dependent interactions, suggesting a role for more flexible methods such as machine learning techniques and models.

2.3 Mental health in older adults

MH, including depression, anxiety, loneliness, and SOC isolation, strongly shapes quality of life. Evidence from TILDA shows strong links to both physical and SOC factors. [36] found that participation in physical, SOC, and cognitive activities summarised by the Act-Belong-Commit (ABC) indicator was associated with lower depression and anxiety. This suggests SOC integration and lifestyle protect mental wellbeing. However, [37] reported that self reported measures may exaggerate such associations..., raising concerns of bias. These two findings conflict: the effect of SOC engagement is strong when self reported and weaker when objectively assessed.

Objective data strengthen the case for activity. Accelerometer based measures show a dose response link between moderate to vigorous activity and lower

2. LITERATURE REVIEW

depressive symptoms [12]. Yet reverse pathways are also evident: incident falls and functional decline predict later depression, refereed by FOF [34]. Together, these results reveal a bidirectional relationship, MH both influences and is influenced by physical decline. Most studies rely on broad self report scales, such as CES-D or Hospital Anxiety and Depression Scale (HADS), with little biomarker or clinical validation.

This literature suggests MH in ageing cannot be modelled as a one way effect. It requires longitudinal, multidimensional approaches that can capture mutual pathways, an area where machine learning is well suited, especially when combined with probabilistic models that estimate the likelihood and posterior probability of MH outcomes given multiple interacting features.

For example, we can estimate:

$$P(\text{Depression} \mid \text{Activity, Falls, SOC Engagement}),$$

quantifying how these interacting features together influence MH outcomes. Probabilistic modelling helps determine which features have the strongest effect and which may be redundant for predictive purposes.

2.4 Nutrition and biological ageing

Nutrition interacts with both biological and psychosocial systems, shaping ageing trajectories and FR. Biomarkers like vitamin D, lutein, and zeaxanthin have been linked to physical and cognitive outcomes in TILDA. [11] showed that higher plasma lutein and zeaxanthin dietary carotenoids found in leafy vegetables were linked to slower FR progression. By contrast, [38] found vitamin D deficiency predicted DM, musculoskeletal decline, especially among adults with low sunlight exposure. These studies point to different nutritional mechanisms: antioxidant protection versus metabolic regulation.

2.4.1 Bayesian perspective on nutritional biomarkers

The effect of nutrition on ageing outcomes can be formalised probabilistically using Bayes theorem:

$$P(\text{Outcome} \mid \text{Nutritional Status}) = \frac{P(\text{Nutritional Status} \mid \text{Outcome})P(\text{Outcome})}{P(\text{Nutritional Status})},$$

where the likelihood function $P(\text{Nutritional Status} \mid \text{Outcome})$ quantifies how observed nutritional biomarkers relate to health states, and the posterior $P(\text{Outcome} \mid \text{Nutritional Status})$ captures updated beliefs about outcomes, including uncertainty in small sample studies. This helps identify which biomarkers are most informative for predicting ageing outcomes which supports feature selection in ML models.

2.4.2 Contextual and integrative considerations

Malnutrition also reflects SOC and functional risks. [27] found that hospitalisation, functional loss, and poor SOC support increased malnutrition risk, with notable sex differences. This highlights a limitation in much nutritional research: biological markers are studied without integrating SOC context.

The difference between biological and chronological age is a major point of debate. Biomarker based metrics that better capture these cumulative exposures, like FR indices or epigenetic clocks, frequently vary from chronological age. McCarthy and Azizi link metabolic syndrome and nutritional deficits to accelerated biological ageing [39, 40]. Although the majority of the evidence is cross sectional and has little ability to capture changes over time, these indicators offer a more precise prediction of vulnerability. Reliability is also decreased in biomarker research with small sample sizes, which can be formalised through the likelihood function $P(\text{Data} \mid \theta)$ and its impact on posterior uncertainty in Bayesian inference.

Taken together, nutrition interacts with biological, metabolic, and SOC systems. Isolating single nutrients risks oversimplification. Integrative modelling that combines biomarkers, clinical factors, and SOC determinants supported by machine learning pipelines offers a more promising route. Models that estimate

2. LITERATURE REVIEW

the joint distribution of features and outcomes can better capture the complex dependencies relevant to biological ageing.

2.5 Machine Learning in ageing and chronic disease

ML is increasingly applied in ageing research because it can capture non linear and multidimensional relationships that traditional regression models, such as logistic regression, often miss [22, 23].

2.5.1 Feature sets in ML studies

Studies using ML in ageing research include many types of predictors: clinical measures, sociodemographic factors, lifestyle habits, cognitive tests, and psychosocial variables. A recurring pattern, however, is that models often underline easily measured physical health data, such as comorbidity counts, mobility scores, or vital signs, while more complex but important predictors, including nutrition and SOC factors, are underrepresented [32]. This imbalance may limit the ability of models to capture the full complexity of chronic disease risk with ageing, especially considering that MH and nutrition exert independent and interacting effects on later life outcomes (here). Models that focus only on ready available clinical data risk reproducing the same biases seen in traditional regression approaches.

2.5.2 ML models used

The methodological approaches used in ageing research span both traditional and advanced models. Logistic regression and Markov models remain common baselines due to their interpretability and simplicity, yet they are often unable to capture non linearities or higher order interactions. More recent work has shifted towards ensemble methods such as RF, gradient boosting machines GBM, as well as DNNs, which generally achieve higher predictive accuracy.

Similarly, [41] evaluated mortality prediction using a standard logistic regression model with demographic, clinical, and functional predictors, achieving an

AUC of 0.78, whereas ensemble methods in [22] reached 0.854 for DM prediction, showing how flexible, non linear models can better capture complex interactions among ageing related predictors.

2.5.3 Performance and Validation

Predictive performance across studies is generally moderate to strong, with AUC values between 0.70 and 0.90 for outcomes like DM, UGS, depression, and mortality. This indicates that models can distinguish reasonably well between individuals who will or will not experience an outcome.

However, there are important limitations. Many studies rely only on internal validation, testing models on the same data used for training. This can inflate performance estimates and reduce confidence in applying the model to new populations. To add, most studies are cross sectional, so causal or temporal effects cannot be assessed. When combined with the underuse of nutrition and SOC predictors, these issues indicate that current models may not fully represent the multidimensional nature of ageing and could produce context specific results rather than generalisable predictions [32].

2.5.4 Explainability in ML for ageing

As ML models become more complex, interpretability emerges as a critical challenge. Explainability tools like SHAP can highlight feature importance, but their use is inconsistent, particularly in deep learning models [4, 6, 22]. These methods improve transparency and can highlight unexpected drivers of outcomes, such as psychosocial variables being stronger predictors than biomedical ones, yet black box concerns remain, especially for deep learning models, where feature interactions are difficult to track. For ageing research, where clinical adoption requires both accuracy and trust, explainability is not optional but essential. Current efforts show promise but remain fragmented, with no consensus on best practice. This gap links directly to the broader methodological limits discussed in the next section on gaps in the literature.

2.6 Gaps in the Literature

First, most studies rely on TILDA or similar cohorts of Irish and predominantly older adults (50+). This limits confidence in how well results transfer to other populations. This thesis focuses on TILDA studying ageing in depth but also makes use of the harmonised dataset, which links TILDA with comparable UK and US cohorts. This helps place findings in a broader international context while retaining a core emphasis on older adults.

Second, there is an over reliance on physical health predictors such as comorbidity counts and mobility measures, while nutrition, SOC, religious or spiritual, and environmental variables are often overlooked. As shown in [11], nutritional biomarkers like lutein and zeaxanthin are strong predictors of FR, yet these are rarely integrated into broader ML pipelines. By contrast, studies such as [27] highlight the role of SOC support and functional decline in malnutrition risk. Together, these examples show that excluding nutrition and SOC factors risks producing models that reinforce the same biomedical biases seen in regression heavy work. From a Bayesian viewpoint, the omission of key features reduces the explanatory power of the model and weakens the posterior probability $P(\text{Outcome} \mid \text{All Features})$ compared to $P(\text{Outcome} \mid \text{Limited Features})$. Including a richer set of features increases the likelihood that the model accurately reflects the true data generating process.

Third, methodological choices remain conservative. Regression models remain dominant, likely due to their interpretability and ease of implementation. However, they often fail to capture non linear interactions and higher order dependencies present in ageing related data. Comparisons with more advanced approaches such as DNNs or transformers are rarely explored. For instance, [41] achieved an AUC of 0.78 using logistic regression for mortality prediction in TILDA, whereas [22] reported an AUC of 0.854 with ensemble methods for DM incidence. This contrast illustrates that more flexible ML models can provide superior predictive performance while maintaining interpretability when combined with explainability techniques. Where else ensembles or brainPAD approaches have been attempted, validation is almost always internal, with little cross cohort testing. Weak validation and limited feature sets reduce the generalisability of the

estimated conditional probabilities and increase the risk of overfitting. Maximum likelihood estimation and Bayesian inference can provide a formal framework for assessing model fit and updating predictions as more diverse and representative data become available.

Fourth, many outcomes relevant to older adults such as pain, sleep quality, and subjective wellbeing remain underexplored. For example, [11] focused on plasma biomarkers for FR without assessing sleep or pain, and UGS models such as [6] similarly omit subjective quality of life. Omitting these outcomes limits the real world application of predictive models. Including broader outcomes in ML pipelines can better capture holistic ageing trajectories and improve interpretability for interventions targeting older adults.

Fifth, cross sectional bias and data handling weaken reliability. Much nutritional work, such as [11], relies on single time point biomarker data, which cannot capture longitudinal change. By contrast, studies applying multi wave TILDA data [26] demonstrate the added value of modelling temporal dynamics. Similarly, missing data are often handled through listwise deletion, as in [41], which risks biasing results toward healthier participants, whereas more robust imputation approaches remain underused. Missing or poorly handled data can distort the likelihood function and hence the posterior predictions, reducing confidence in the model’s outputs.

Finally, interpretability challenges persist. While tools like SHAP have been applied to highlight feature importance [4, 22], their use is inconsistent, and consensus on best practice has yet to emerge. Without transparent explanations, even accurate models may face barriers to clinical acceptance. Taken together, these examples show that ageing research continues to rely on narrow predictors, conservative models, and weak validation. Addressing these gaps requires integrative, diverse, and transparent approaches that link multidimensional predictors with explainable ML.

2.7 Chapter Summary

This chapter reviewed evidence on ageing and chronic disease across three domains, FR, physical function, MH, nutrition, and biological ageing. We also

2. LITERATURE REVIEW

examined how machine learning has been applied to predict later life outcomes. The findings showed that while FR is partly reversible, it remains strongly tied to mortality, that MH both shapes and is shaped by physical decline, and that nutrition interacts with biological, metabolic, and SOC systems in ways that are often underestimated.

While machine learning has improved prediction accuracy in ageing research, challenges remain in integrating diverse predictors and ensuring interpretability and longitudinal validation.

Addressing these challenges requires the integrative, longitudinal, and transparent approaches that combine diverse predictors. These gaps motivated the feature engineering and modelling strategies implemented in Chapters 3 and 4, including the use of harmonised variables to ensure reproducibility and generalisability, for ML modelling.

Chapter 3

Technical Approach

3.1 Study Design

TILDA provides longitudinal data across five waves (Waves 1–5) for community dwelling adults in Ireland. For contextual comparison and benchmarking against international cohorts see Chapter 1. An external, harmonised TILDA dataset, was also inspected and prepared. The harmonised dataset, which has been pre aligned to international standards e.g., the Gateway to Global Ageing Data (G2G), contains a subset of variables mapped to a common cross study format and thus differs from the raw TILDA files. Although the harmonised dataset was simpler to process, it was still subjected to the variable filtering and numerical standardisation steps developed as part of this research to ensure compatibility with the five wave cleaned datasets prior to modelling.

3.2 Cohort Definition

All participants who provided data in at least one TILDA wave were considered eligible. To maximise the available sample for machine learning, no exclusions were applied on the basis of age, medical conditions, mobility, cognitive status, or health centre attendance.

Participants were removed only in the rare case that, following manual variable filtering, no analytically usable variables remained for that individual. Filtering

3. TECHNICAL APPROACH

occurred primarily at the *variable level* during the manual screening stage, which involved features removed for extreme missingness, lack of variation, or non informative structure. This approach preserves the largest possible sample while ensuring all retained inputs are analytically meaningful for downstream analysis and model development.

- **Participant inclusion:** All respondents who contributed data in at least one wave were included in the initial sample.
- **Usable data requirement:** Participants were excluded only if all retained variables were missing for that individual after manual variable screening.
- **Longitudinal retention:** Individuals were retained regardless of how many waves they contributed to, enabling initial maximum dataset utilisation.
- **No clinical exclusions:** No participant was removed due to medical diagnoses, cognitive scores, functional limitations, or non attendance at the health centre assessment.

This approach prioritises data integrity and sample size for machine learning applications, differing from exclusion-heavy cohort definitions often used in traditional epidemiological studies [42].

3.3 TILDA Data Collection and Screening

TILDA data collection occurred over five longitudinal waves as detailed in Table 3.1.

Each wave included three primary components:

1. **Computer Assisted Personal Interview (CAPI):** Home based, computer assisted personal interviews covering health, socio demographics, and family.
2. **Self Completion Questionnaire (SCQ):** Paper based self completion questionnaires capturing sensitive information on mental health, lifestyle, and wellbeing.

3.3 TILDA Data Collection and Screening

3. **Health Assessment:** Clinical and biomarker measurements conducted in dedicated centres or at home (Wave 1 and Wave 3).

All TILDA data were pseudonymised prior to release, meaning that personal identifiers such as names and addresses were removed or replaced with unique codes to protect participant confidentiality. Each participant received a unique identifier ('id'), along with household and cluster identifiers ('household', 'cluster'), allowing longitudinal linkage across waves without revealing individual identities.

Pseudonymisation ensures that analyses comply with data protection standards while retaining the structural and analytical integrity of the dataset. From a data cleaning and machine learning perspective, pseudonymised identifiers enable accurate merging of data across waves, preserve the longitudinal design, and support reproducible preprocessing and feature engineering. While these identifiers cannot be used for analyses of personal demographics like names or addresses, this limitation does not affect the modelling of health outcomes, lifestyle factors, or other variables of interest.

Prior to any data cleaning or modelling, a structured multi stage workflow was implemented to screen and reduce the dimensionality of the raw TILDA datasets. This workflow separates automated diagnostic profiling from manual variable exclusion and ensures transparency in all variable selection decisions.

The process consisted of the following steps:

1. **Raw data loading:** Raw TILDA PMF files were loaded without cleaning or recoding. Waves 1–5 were loaded from Statistical Package for the Social Sciences (SPSS) (.sav) files, while the harmonised dataset was loaded from a (Stata) (.dta) file due to technical compatibility issues.
2. **Automated variable profiling (Python):** For every variable, summary statistics were computed programmatically, including data type, proportion of missing values, number of unique values, most frequent value, and frequency of the most frequent value. These summaries were exported to spreadsheet format for inspection.

3. TECHNICAL APPROACH

3. **Manual variable screening (spreadsheet based):** Using predefined rule based thresholds, variables were manually reviewed and flagged for exclusion based on missingness, lack of variation, dominance of a single category, or administrative function.
4. **Creation of retained variable lists:** Variables not flagged for exclusion were programmatically concatenated in `.xlsx` and manually transferred into wave specific JavaScript Object Notation (JSON) files representing the retained variables of interest.
5. **Reloading filtered datasets:** The JSON files were then used to subset the original raw datasets, producing reduced datasets with identical rows but fewer variables. No cleaning, recoding, or imputation was applied at this stage.

These reduced datasets form the direct input to the automated preprocessing and modelling pipeline described in Chapter 4.

Table 3.1: Dimensions of raw TILDA and harmonised datasets (prior to filtering).

Dataset	Collection Period	Participants (N)	Raw Variables (N)
Wave 1	2009–2011	8501	1619
Wave 2	2012–2013	7206	724
Wave 3	2014–2015	6397	1029
Wave 4	2016	5713	990
Wave 5	2018–2019	4978	983
Harmonised	2016 (release)	8501	572

Following the five step workflow outlined above, Python scripts were used to profile all variables across each wave. For every variable, the following metrics were computed: data type, number of missing values, number of unique values, most frequent value, and frequency of the most frequent value. These summaries were exported to spreadsheet format to support structured manual review prior to variable dropping.

Manual filtering was applied according to the following criteria:

3.3 TILDA Data Collection and Screening

- **Missingness:** Drop variables with $> 80\%$ missing values. This threshold balances retaining sufficient feature coverage with ensuring statistical reliability. Variables exceeding this threshold provide limited information in most cases, increase imputation uncertainty, and reduce effective sample size, which is particularly critical in longitudinal machine learning analyses.
- **Unique values:** Drop variables with a single observed value, as they contain no discriminative information and cannot contribute to modelling or inference. Variables with low cardinality (2–4 unique values) were retained for review, as they frequently represent meaningful categorical responses (e.g., binary indicators or ordinal scales) relevant to ageing outcomes.
- **Dominant category (top frequency):** Variables where a single category (sample value) occurs in over $> 90\%$ of observations were flagged. Such dominance indicates limited variability, reducing analytical utility and predictive value. Variables exceeding this threshold were marked for review or exclusion.
- **Identifiers / administrative fields:** IDs, version tags, and sampling weights were identified using the PMF metadata. Variable names and descriptions were exported from each wave’s PMF file into CSV files (code provided in Appendix A) to systematically document and review features prior to filtering. These exports enabled structured manual inspection and traceability between raw PMF metadata and downstream feature selection decisions.

As part of the manual screening and filtering workflow. Variables exceeding exclusion thresholds (e.g., high missingness, single value, dominant category, administrative identifiers) were visually flagged (e.g., marked in red). Non flagged variables were programmatically concatenated using spreadsheet formulas (e.g., `=TEXTJOIN(", ", TRUE, J2:Jn)`) and then manually transferred into wave specific JSON files for each wave. These files represent the variables of interest retained after manual screening and filtering and serve as inputs for the upcoming automated data preprocessing & cleaning in Chapter 4.

3. TECHNICAL APPROACH

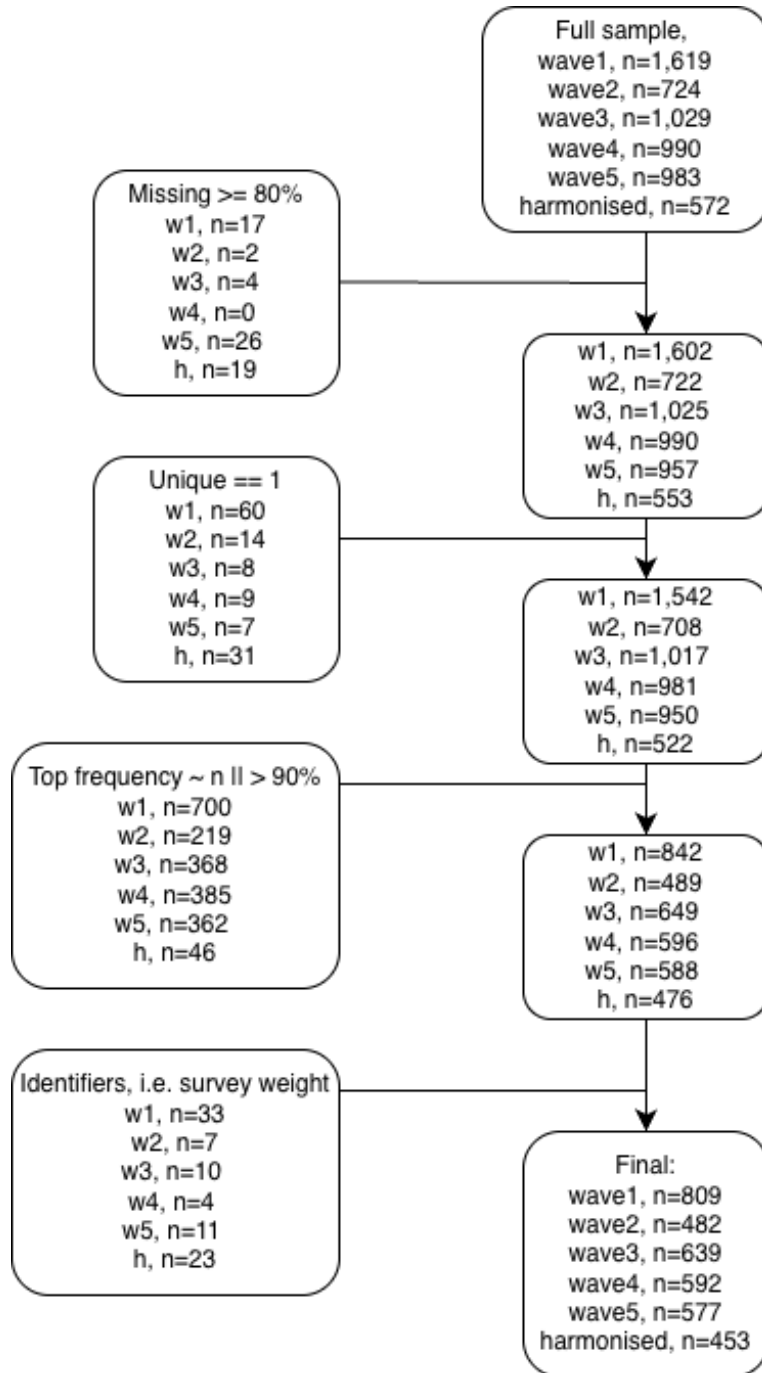


Figure 3.1: Flowchart of the five step workflow for variable screening and filtering. Placeholder counts indicate the progression of remaining variables per dataset from raw to filtered datasets after each step, ready for automated preprocessing.

The retained variables, defined in JSON files, were subsequently used to subset the raw datasets. This produced reduced CSV datasets per wave, containing only the variables of interest. At this stage, no cleaning or recoding was performed. The operation served purely as dimensionality reduction, preparing the data for the automated nine-scenario cleaning pipeline described in Chapter 4.

Importantly, all variable exclusion decisions were completed at this stage. Subsequent preprocessing steps focus exclusively on numerical standardisation, recoding, and preparation for machine learning rather than further feature removal. With the variables of interest selected and the datasets reduced to a manageable dimensionality, the next stage focuses on automated preprocessing, recoding, and preparation for machine learning.

3.4 Thematic Grouping

After manual filtering, the retained variables were organised into six thematic domains based on ageing literature and study objectives. Each variable’s prefix, as defined in the PMF metadata (Table 3.2), indicates its source or research domain. This systematic assignment ensures that each thematic group contains conceptually related features and provides a clear structure for downstream automated preprocessing and machine learning.

Each variable in TILDA was assigned a prefix indicating its data source or research domain. Home based CAPI variables use a two letter module code followed by a three digit question code (e.g., `ph121`). Variables with multiple responses or loops include an underscore and number to specify the response or iteration (e.g., `ph201_05`).

Variables from the self completion questionnaire are prefixed with SCQ, while derived variables constructed from either CAPI or SCQ data use uppercase codes corresponding to their research area (Table 3.2). These prefixes provide a consistent framework for categorising features and guiding the assignment into thematic domains for downstream analysis.

Using the PMF prefixes as a guide, variables were assigned to six thematic groups (Table 3.3), each capturing a conceptually coherent domain: physical,

3. TECHNICAL APPROACH

Table 3.2: PMF Derived variable prefixes [3].

Variable Prefix	Research Area
ADL	Activities of daily living
BEH	Health behaviours
BLOODS	Results of blood sample analysis
BP	Blood pressure from sphygmomanometer
COG	Scales and questions on cognition
CRT	Choice reaction time (cognitive test)
DIS	Disability, functional impairment, helpers
FR	Frailty, gait, exercise, falls and fractures
G2G	Gateway to Global Aging Data
GRIP	Grip strength
IADL	Instrumental activities of daily living
INC	Income variables
MD	Medication data
MH	Scales and questions on mental health and wellbeing
SCR	Screening tests
SES	Social class and socio
SOC	Social

cognitive, social, and behavioural aspects of ageing. This structure supports reproducible preprocessing and helps machine learning models focus on meaningful, longitudinally robust features.

Specifically, the thematic grouping:

1. Provides a clear framework for automated cleaning and filtering rules.
2. Ensures that model input variables are conceptually aligned with study objectives.

The six thematic groups integrate information from CAPI, SCQ, and derived variables, covering socio demographic context, physical health, functional status, lifestyle behaviours, nutrition/metabolic biomarkers, and psychological/cognitive measures. This structured grouping enhances interpretability, ensures comprehensive domain coverage, and provides a foundation for downstream predictive modelling and risk assessment.

3. TECHNICAL APPROACH

Table 3.3: Thematic grouping of longitudinal features.

ID	Thematic Group	Description	Representative Variables
1	Socio Demographic & Context	Baseline Stratification Factors	Age
			Gender
			Education
			Income Variables (INC)
			Marital Status
			SOC
2	Physical Health & Clinical Baseline	Self reported and clinical indicators	Self Rated Health
			Screening Tests (SCR)
			BMI
			BP
			Comorbidity Count Medication Data (MD)
3	Functional Status & FR	Measures of mobility and physical decline	FR Index
			UGS
			Falls History
			Instrumental Activities of Daily Living (IADL)
			Disability, Functional Impairment and Helpers (DIS)
4	Lifestyle & Behavioural	Modifiable health related behaviours	Smoking
			Alcohol Intake
			PA
			International Physical Activity Questionnaire (IPAQ)
5	Nutrition & Metabolic Biomarkers	Objective nutritional and metabolic markers	Plasma Carotenoids
			Lutein
			Zeaxanthin
			Vitamin D/B12
			Cholesterol
6	Psychological & Cognitive	Mental health and cognitive performance	CES-D
			Loneliness
			Scales and Questions on Cognition (COG)
			Choice Reaction Time (Cognitive Test) (CRT)

Chapter 4

Implementation Details

Following manual variable filtering in (Chapter 3), the reduced datasets, containing only the variables of interest. To further understand the characteristics of these retained variables, the datasets were profiled using an exploratory visualiser (`visualiser.ipynb`) implemented with `pandas`, `numpy`, `matplotlib`, and `seaborn`. This step examined data types, missingness patterns, unique value distributions, and representative sample values across all waves, alongside an automated scenario detection audit. The purpose of this profiling stage was not transformation, but to expose structural inconsistencies and latent formatting issues that would undermine naive numerical conversion.

Table 4.1: Input shape for cleaning pipeline (post manual variable filtering).

Dataset	Row (participants)	Column (features)
Wave 1	8501	809
Wave 2	7206	482
Wave 3	6397	639
Wave 4	5713	592
Wave 5	4978	577
Harmonised	8501	453

For example, in Wave 1, the visualiser detected 650 categorical columns, 150 (`float64`) columns, and 9 object columns, while the harmonised dataset contained 304 categorical columns, 148 `float32` columns, and several integer columns.

4. IMPLEMENTATION DETAILS

Age distributions confirmed that all participants were 50+ across waves, and missing value summaries indicated substantial variation (e.g., mean missing values ranging from 4.7% in Wave 4 to 35% in the harmonised dataset, with some variables exceeding 77% missing).

Despite this, the exploratory visualiser was limited because the raw data were not yet cleaned, many outputs were incorrect or nonsensical due to hybrid formats and inconsistent coding. For instance, inspection revealed that many variables across both the wave specific and harmonised datasets, while appearing numeric at first glance, in fact contained hybrid representations combining numeric prefixes with descriptive text (e.g., “1.Rarely or none of the time (less than 1 day)”, “4.All of the time (5–7 days)”, or prefixed labels such as “1.male”, “2.female”). A naive string stripping approach would therefore produce incorrect numeric values (for example, reducing “5–7 days” to 5) resulting in silently introducing semantic distortion. Similarly, numeric columns sometimes contained sentinel codes (e.g., -98, -99) or embedded symbols (‘\$’, ‘cm’, ‘%’) that skewed visual summaries.

These insights, revealed through systematic profiling, guided the development of the subsequent cleaning pipeline, ensuring consistent numeric logic representations (e.g., midpoints for ranges, standardised sentinel handling).

4.1 Cleaning Engine

The core challenge in converting heterogeneous variable types across the six datasets (Table 4.1) was achieving a single numerical standard suitable for downstream analysis. This required enforcing *numerical homogeneity*, defined here as representing all retained features as consistent numerical values (`float64`) with a unified missing or non response sentinel (`-1.0`). To address this, the solution was implemented as a two stage system. The first stage performs automated *variable scenario detection*, identifying the structural form of each variable based on observed value patterns using the `detect_column_scenario_mapping` function (Appendix C). This stage includes additional diagnostic categories used during development to surface unhandled edge cases.

The second stage applies a deterministic *six step numerical cleaning engine* (Appendix D). While nine scenario categories were initially defined during devel-

opment, three served exclusively as validation fallbacks and were progressively eliminated as detection coverage improved. The final pipeline therefore consists of six active transformation steps, applied sequentially and uniformly across all waves.

4.1.1 Stage 1: Variable scenario mapping and validation

Stage 1 focused on automatically identifying how each variable should be handled during cleaning. An automated detection function (`detect_column_scenario_mapping` in Appendix C) was developed to classify variables based on their observed value structure, including the presence of sentinel codes, numeric annotations, ordinal labels, and mixed string numeric formats. While multiple intermediate detection categories were used during development, these served solely as internal routing logic. While all variables passed sequentially through the six step cleaning pipeline, most were fully resolved in the earlier steps (Steps 1–5). The final step (Step 6) was applied only to residual variables that could not be deterministically resolved by earlier transformations.

The pipeline’s first stage utilised an automated detection function (Appendix C), powered by the `re` (Regex) Python library, to categorise variables based on observable value patterns. During development, additional detection categories (e.g., “single value categorical”, “special symbol mixed”, and “unknown/complex”) were intentionally included as diagnostic fallbacks to surface unhandled edge cases due to the complexity and size of the data. As detection rules were continuously refined and coverage improved, variables initially routed to these fallback categories were progressively absorbed into earlier, well defined scenarios, resulting in zero variables remaining in these categories for the final pipeline execution.

To ensure robustness, the classification output was immediately processed by a validation routine that generated a comprehensive dictionary of feature states before and after cleaning, including representative sample values for each variable. This allowed direct inspection of how the same variable was represented across different waves prior to cleaning, particularly in cases where identical semantic categories appeared in differing orders or formats. By comparing these grouped

4. IMPLEMENTATION DETAILS

samples before and after each cleaning step, classification errors and inconsistent mappings were identified and corrected prior to execution of subsequent transformations.

The dictionary generated inside the “show_grouped_vars_for_scenarios” function was formatted to create an automatic friendly “.md” file for rapid, surface level verification, which honed the final mapping dictionary D for all features across all waves. This process of using automated detection followed by visual, manual validation was essential for correcting errors in classification logic before the subsequent scenario transformations were executed in the pipeline.

4.1.2 Stage 2: Deterministic six step cleaning sequence

The nine input scenarios were processed by the *six step cleaning engine*. Resolving simple and unambiguous cases first increased the proportion of clean numerical features before addressing more complex structures, particularly censored and multi class scenarios (Figure 4.1). This deterministic sequence applied the same logic across all waves to ensure consistent data alignment (see Appendix D).

A key difficulty in cleaning the five TILDA waves was the lack of guaranteed ordering in categorical values for the same variable across different waves. Table 4.2 shows the positional location for each categorical variable in Wave 4 and Wave 5. If the cleaning engine relied on the positional index for transformation, the inconsistent categorical ordering would result in different numerical codes for the same semantic category across waves. To overcome this, the cleaning engine was designed to be *order invariant* wherever possible, particularly in the later steps (see Table X - where table X is a version of the CGPT generated one below with the code value replacing ‘Pos’ column and the Wave categories aligned (ie only three columns in total)).

This issue became most pronounced in the final cleaning stage (Step 6), which addressed multi-class and format variables. These variables frequently combined ordinal labels, frequency expressions, numeric ranges, and free text descriptors, with category ordering differing across waves. Encoding such variables using positional or alphabetical rules would therefore result in inconsistent numerical representations for semantically identical responses.

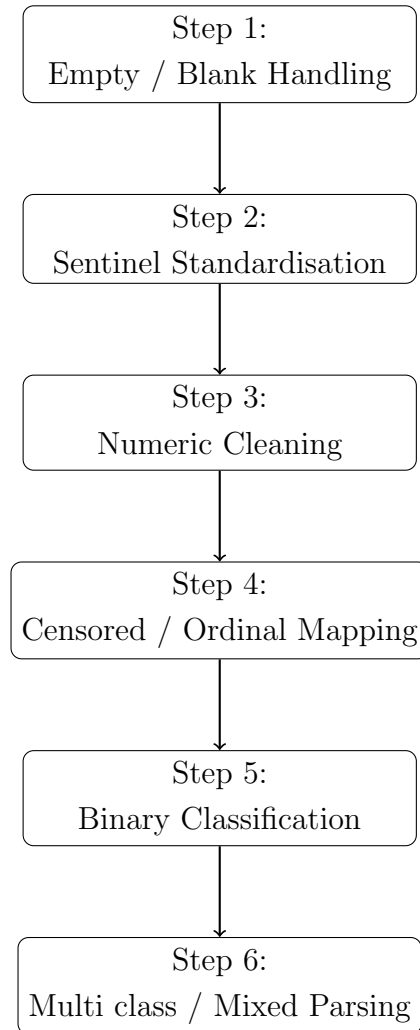


Figure 4.1: Flowchart of the final six step cleaning pipeline applied to all TILDA waves. See full transformation tables in Appendix B.7 and D for detailed scenario progression code at each step.

4. IMPLEMENTATION DETAILS

The final transformation step (Step 6) therefore addresses residual multi-class and mixed-format categorical variables that cannot be deterministically resolved by earlier steps. Importantly, this step does not reuse mappings learned from previous waves. Instead, variables are cleaned dynamically based on their observed value structure within each execution, while enforcing consistent semantic interpretation across waves.

For example, gender variables appeared under different names and value encodings across datasets (e.g., `sex`, `gd002`, `ragender`), with category ordering and numeric prefixes varying by wave. Values such as `['Male', 'Female']`, `['Female', 'Male']`, and prefixed forms such as `['1.male', '2.female']` were all mapped to a consistent numerical representation (1.0 for male, 0.0 for female), independent of ordering or source dataset.

Similarly, multi-class variables such as insurance status (`ins3`) were dynamically parsed from textual categories into stable numeric identifiers, with non-response values consistently mapped to the sentinel value `-1.0`. This ensured that semantic equivalence, rather than positional ordering or wave-specific encoding, governed numerical assignment.

For multi-class scenarios (Step 6) also the final step, consistency was achieved by constructing explicit mapping dictionaries derived from grouped category strings rather than positional ordering. These mappings were informed by iterative inspection of scenario grouped sample outputs, rendered in human readable `.md` format using the `show_grouped_vars_for_scenarios` function. This ensured that semantically identical response categories were assigned stable numerical values across waves, regardless of their original ordering or survey specific presentation.

4.1.3 Scenario steps and their technical focus

1. **Empty / Blank Values:** Converts all NaN, empty strings, and whitespace to the standard non response value `-1.0`.
2. **Sentinel Values:** Standardises wave specific non response codes (e.g., `-98`, `-99`) and documented textual sentinels (e.g., `"dk"`, `"not applicable"`) from the TILDA PMF release guide to the unified `-1.0` placeholder.

Table 4.2: Inconsistent category ordering for the same feature *ph009* across waves and its impact on positional (order-based) label encoding.

Pos	Wave 4		Wave 5	
	Category	Code	Category	Code
1	Good	3.0	Very Good	4.0
2	Fair	2.0	Fair	2.0
3	Poor	1.0	Excellent	5.0
4	Excellent	5.0	Good	3.0
5	DK	-1.0	Poor	1.0
6	Very Good	4.0	DK	-1.0
7	RF	-1.0	RF	-1.0

3. **Numeric Transformation:** Removes non numeric characters (units, symbols, white spaces) and converts censored or annotated numeric strings (e.g., "50+", "<10", "1-2.5") to a numerical float equivalent (e.g., boundary value or midpoint).
4. **Censored Categorical:** Applies explicit, known ordinal transformations and predefined mappings to features where text labels represent a latent numerical or ordered scale (e.g., health rating scales, Likert ranges). This step was intentionally performed manually for a subset of low count features to build intuition and reliable mapping dictionaries for the more complex subsequent scenarios. This included conversions for calendar based frequency strings like "twice a week" to a numerical weekly equivalent.
5. **Binary Categorical:** Automatically maps two level features (e.g., Yes/No, Male/Female) to 1.0/0.0. Importantly, this mapping is *order invariant*: the numerical assignment does not depend on the original ordering of category labels in the raw data (e.g., [Male, Female] versus [Female, Male]). Instead, the encoding relies on *semantic detection rules* (e.g., assigning 0.0 to negated or absence indicating strings such as "no", "not", or wave specific refusal codes, under context constraints) alongside *count based detec-*

4. IMPLEMENTATION DETAILS

tion rules (where a variable exhibits exactly two non missing unique values). This design choice ensures consistent binary representation across waves and features, independent of label ordering or survey specific value presentation.

6. **Multi-class / Mixed (The Complex Scenario):** This final step processes all remaining multi label, mixed type, and string numeric hybrid columns. The challenge was made more complex by the fact that the order of categorical string values could vary across waves, as discussed previously. To handle this, numerical assignments were made using an explicit dictionary lookup based on the original order of the sample values , rather than relying on positional or alphabetical order. This ensured consistent numerical representation. The step performs extensive lexical processing, dedicated number extraction, and rule based mapping to convert dense, un-standardised text into numerical floats, ensuring all residual variable types are successfully quantified. The transformed values were subject to increased manual verification due to the variable complexity.

4.2 Cleaning Validation

The robustness of the automated cleaning pipeline was evaluated in two stages. First, the internal consistency of cleaned predictors was assessed against established biological relationships. Second, full model replication was performed using a secondary study to confirm predictive utility.

4.2.1 Pairwise correlation validation

Correlation analysis was conducted on the seven primary predictors identified by [4] as explaining 95% of the peak (GSR) model performance, which is defined in this study as the difference between maximum and usual gait speed. Since the GSR outcome itself is not natively available in the public Wave 3 dataset, validation was focused on the directional relationships between the predictors themselves. This internal consistency check ensures that the biological and clinical signals reported in the literature are preserved within the dataset (post cleaning).

Figure 4.2 displays the correlation matrix for the selected predictors. These top 7 predictors were used to establish directional relationships for verifying that the numerical parsing pipeline specifically the handling of sentinel values was successful.

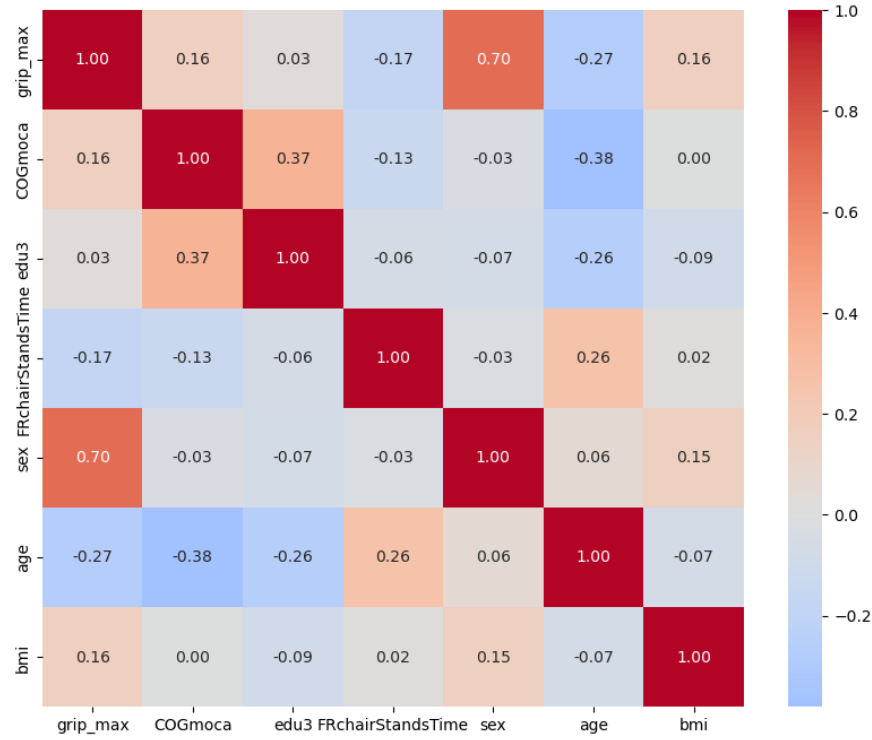


Figure 4.2: Correlation heatmap for top 7 predictors (Wave 3).

4.2.2 Model Validation

Full model validation was performed by replicating the methodology of [16], which predicts 8-year mortality across Waves 1–5 using baseline Wave 1 predictors. The replication of AUC scores in the range of 0.70–0.90 serves as a functional validation.

4. IMPLEMENTATION DETAILS

Table 4.3: Directional validation of the seven features explaining 95% of GSR variance as identified by [4].

Predictor 1	Predictor 2	Expected Direction / Justification	Observed r
Grip Strength (GRIP)	Age	Negative (-) ; GRIP is a positive predictor of GSR, while age is a negative predictor [4].	-0.265
GRIP	Chair Stands Time	Negative (-) ; Higher strength (grip) is expected to correlate with faster (lower) performance time in physical tests [4].	-0.170
Montreal Cognitive Assessment (MOCA) Score	Age	Negative (-) ; Cognitive performance (MOCA) contributes positively to GSR, whereas increasing age is associated with reduced reserve [4].	-0.381
Education	MOCA Score	Positive (+) ; Third level education and cognitive scores are both cited as positive contributors to cognitive reserve and GSR [4].	0.367
BMI	Age	Weak / Mixed ; BMI and age both negatively impact GSR, though their internal correlation in older cohorts is typically weak [4].	-0.073

Chapter 5

Empirical Evaluation

You can also use this kind of referencing if needed in the text:

Chapter 6

Conclusion

6.1 Summary

6.2 Conclusions

6.2.1 Data availability Statement

The data analysed in this study is subject to the following licenses/ restrictions:
The datasets generated during and/or analysed during the current study are not publicly available due to data protection regulations but are accessible at TILDA on reasonable request. Requests to access these datasets should be directed to <https://tilda.tcd.ie/data/accessing-data/>.

6.2.2 Ethics Statement

The studies involving human participants were reviewed and approved by Faculty of Health Sciences Research Ethics Committee at Trinity College Dublin, Ireland. The patients/ participants provided their written informed consent to participate in this study.

6.3 Contributions

6.4 Future Work

Appendix A

Python code for exporting PMF metadata to CSV

```
1 import pandas as pd
2 import pyreadstat
3 from pathlib import Path
4 import json
5
6 data_files = {
7     "wave1.json": "data/0053-01_TILDA_Wave1_2009-2011/0053-01
8         _TILDA_Wave1_Data/SPSS/0053-01_TILDA_Wave1_PMF_v1.12.sav",
9     "wave2.json": "data/0053-01_TILDA_Wave2_2012-2013/Data
10         /0053-02_TILDA_Wave2_PMF_v2.7.sav",
11     "wave3.json": "data/0053-01_TILDA_Wave3_2014-2015/Data
12         /0053-04_TILDA_Wave3_PMF_v3.6.sav",
13     "wave4.json": "data/0053-01_TILDA_Wave4_2016/Data/0053-05
14         _TILDA_Wave4_PMF_v4.4.sav",
15     "wave5.json": "data/0053-01_TILDA_Wave5_2018/Data/0053-06
16         _TILDA_Wave5_PMF_v5.5.sav",
17     "harmonized.json": "data/0053-06_TILDA_Harmonized_2016/Data
18         /0053-03_TILDA_Harmonized_v1.2.dta",
19 }
20
21 def load_df_with_var_of_interest(
22     var_dir="var_of_interest", data_files=data_files, verbose=
23     False
```

```

18 ):
19     dfs = {}
20     var_dir = Path(var_dir)
21
22     for json_name, data_path in data_files.items():
23         file = var_dir / json_name
24         with open(file, "r") as f:
25             vars_list = json.load(f).get("variables_of_interest")
26             if verbose:
27                 print(f"Variables_of_interest_from_{file.name}:_{vars_list[:10]}_...")
28
29             if data_path.endswith(".sav"):
30                 df_full, meta = pyreadstat.read_sav(
31                     data_path, encoding="latin1", apply_value_formats
32                     =True
33                 )
34                 df, _ = pyreadstat.read_sav(
35                     data_path,
36                     usecols=vars_list,
37                     encoding="latin1",
38                     apply_value_formats=True,
39                 )
40             else:
41                 df_full = pd.read_stata(data_path,
42                                         convert_categoricals=True)
43                 df = pd.read_stata(data_path, convert_categoricals=
44                                     True)[vars_list]
45
46             if verbose:
47                 print(
48                     f>Data_shape_(Original):_{df_full.shape[0]}_rows_
49                     x_{df_full.shape[1]}_columns"
50                 )
51                 print(f>Data_shape_(Cleaned):_{df.shape[0]}_rows_x_{
52                     df.shape[1]}_columns")
53                 print("
54                     -----")
55
56     dfs[file.stem] = df

```

Appendix

```
52     return dfs
53
54
55 if __name__ == "__main__":
56     for json_name, path_str in data_files.items():
57         path = Path(path_str)
58         wave_name = json_name.replace(".json", "")
59         if path_str.endswith(".sav"):
60             # SPSS
61             df, meta = pyreadstat.read_sav(
62                 path, encoding="latin1", apply_value_formats=True
63             )
64
65         else: # Stata .dta file
66             df, meta = pyreadstat.read_dta(path, encoding="latin1")
67
68         pd.DataFrame(
69             {"variable": meta.column_names, "label": meta.
70              column_labels}
71         ).to_csv(Path("output") / f"{wave_name}
72                  _variable_descriptions.csv", index=False)
73
74         print(f"-----wave_{path}-----")
75         print(df.shape)
76         print(df.columns[:20])
77         print(
78             "Variable/Feature_descriptions, csv_file_saved_
79             successfully->dir=output/"
80         )
81         df.to_csv((Path("output") / f"{wave_name}.csv"), index=
82                   False)
83
84     dfs = load_df_with_var_of_interest(verbose=True)
```

Appendix B

Scenario Cleaning Pipeline Tables

Table B.1: Initial Raw Feature Counts Categorised by Nine Scenarios (Pre Cleaning)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	9	0	7	0	9	0
Numeric with sentinel	187	59	143	60	100	141
Clean numeric	14	27	18	22	20	8
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	248	159	177	220	207	187
Multi class categorical / mixed	267	166	203	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total Columns	809	482	639	592	577	453

Appendix

Table B.2: Scenario Count Progression: Post Step 1 (Empty / Blank Handling)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	81	31	42	32	75	15
Clean numeric	124	55	123	50	54	134
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Table B.3: Scenario Count Progression: Post Step 2 (Sentinel Standardisation)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	205	86	165	82	129	149
Censored / bracketed numeric	84	71	91	85	85	73
Binary categorical	251	159	179	220	207	187
Multi class categorical / mixed	269	166	204	205	156	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Table B.4: Scenario Count Progression: Post Step 3 (Numeric Cleaning)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	263	117	215	124	172	149
Censored / bracketed numeric	16	23	17	16	17	73
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	227	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Table B.5: Scenario Count Progression: Post Step 4 (Censored / Ordinal Mapping)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	279	140	233	140	189	222
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	251	160	180	221	207	187
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Table B.6: Scenario Count Progression: Post Step 5 (Binary Classification)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	530	300	413	361	396	409
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	279	182	226	231	181	44
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Table B.7: Final Dfs Cleaned: Post Step 6 (Multi-Class / Mixed Parsing)

Scenario	Wave 1	Wave 2	Wave 3	Wave 4	Wave 5	Harmonised
Empty / blank values	0	0	0	0	0	0
Numeric with sentinel	0	0	0	0	0	0
Clean numeric	809	482	639	592	577	453
Censored / bracketed numeric	0	0	0	0	0	0
Binary categorical	0	0	0	0	0	0
Multi class categorical / mixed	0	0	0	0	0	0
Single value categorical	0	0	0	0	0	0
Special symbol / mixed	0	0	0	0	0	0
Unknown / complex mixed	0	0	0	0	0	0
Total	809	482	639	592	577	453

Appendix C

First Stage: Scenario Variable Mapping

```
1 # Sentinel values (numeric and string)
2 # -1 and -1.0 is sentinel but will be used in the clean pipeline
3 standardised_sentinels = {'-1', '-1.0', -1, -1.0} # set ->
    duplicates removed -> counts correct -> single value detection
    is zero
4 numeric_sentinels = [-1, -1.0, -98, -98.0, -99, -99.0,
5                      95, 95.0, 96, 96.0, 98, 98.0, 99, 99.0]
6 string_sentinels = [str(x) for x in numeric_sentinels] + [f"_{x}"
7                  for x in numeric_sentinels] + \
8                  ['na', 'n/a', 'refused', 'missing', 'dk', 'rf', 'refused', "
9                  refuse", 'none', 'no_response', 'nan', 'not_answered', 'no
10                 _answer',
11                 'not_applicable', 'no_consent', 'dont_know', "don't_know", "didn
12                 _t_answer", 'prefer_not_to_say', 'task_not_done', '
13                 insufficient_sample', 'unable_to_record',
14                 '__', '*']
15 blanks = ['', ' ', '\t', '\n']
16
17 SCENARIOS = [
18     "Empty_/blank_values",
19     "Numeric_with_sentinel",
20     "Clean_numeric",
21     "Censored_/bracketed_numeric",
22     "Binary_categorical",
```

```

18     "Multi_class_categorical/_mixed",
19     "Single_value_categorical",
20     "Special_symbol/_mixed",
21     "Unknown/_complex_mixed"
22 ]
23
24
25 # flatten all dtypes across the waves to float and category
26 def flatten_column_types(df):
27     df_flat = df.copy()
28
29     for col in df_flat.columns:
30         if pd.api.types.is_numeric_dtype(df_flat[col]): # Numeric
31             columns
32             df_flat[col] = df_flat[col].astype(float) # Convert
33             to float
34         else:
35             df_flat[col] = ( # Convert to string, strip
36                 whitespace, lowercase, then category
37                 df_flat[col]
38                 .astype(str)
39                 .str.strip()
40                 .str.lower()
41                 .astype('category')
42             )
43
44     return df_flat
45
46 def detect_column_scenario_mapping(df, sample_size=10):
47     summary = []
48
49     # Regex patterns
50     special_symbols_pattern = r'^A-Za-z0-9\s\.,\-\+&/()' #
51         special symbols excluding common numeric formatting
52     bracketed_numeric_pattern = r"\d{1,3}(?:,\d{3})*[-/]\d
53         {1,3}(?:,\d{3})*|\d+\s*[\<>+-]" # patterns like "10-20",
54         "30/40", ">=50"
55     sentinel_set = set(x.lower() for x in string_sentinels) #
56         Lowercase string sentinels for comparison
57
58     for col in df.columns:

```

C. FIRST STAGE: SCENARIO VARIABLE MAPPING

```
52     col_data = df[col].copy()
53     dtype = str(col_data.dtype) # get data type as string
54
55     # Convert to string for sentinel / blank detection
56     str_vals = col_data.astype(str).str.lower().str.strip() #
        lowercase and strip whitespace
57     clean_vals = col_data[~str_vals.isin(sentinel_set) & ~(
        str_vals.isin(blanks))] # remove sentinels and blanks
58
59     # Remove sentinels and blanks for scenario detection only
60     vals_no_sentinel = clean_vals[~clean_vals.astype(str).str
        .lower().isin(sentinel_set)
61                                     & ~clean_vals.astype(str).
        str.lower().isin(blanks)
        ].unique()
62     num_unique_no_sentinel = len(vals_no_sentinel)
63
64     # Check if any sentinel values are present
65     sentinel_present = str_vals.isin(sentinel_set).any()
66
67     # Detection flags
68     empty_detected = str_vals.isin(blanks).any() or len(
        col_data) == 0
69
70     # Numeric detection: coerce to numeric, NaNs for non-
        numeric
71     numeric_vals = pd.to_numeric(clean_vals, errors='coerce')
        # coerce non-numeric to NaN
72     numeric_like = numeric_vals.notna().all() and len(
        clean_vals) > 0 # all values are numeric-like after
        cleaning
73
74     sentinel_detected = str_vals[~str_vals.isin(
        standardised_sentinels)].isin(sentinel_set).any() # or
        col_data.eq(-1.0).any() # check for sentinel presence
75     censored_detected = clean_vals.astype(str).str.contains(
        bracketed_numeric_pattern).any() # check for bracketed
        numeric patterns
76
77     # Binary categorical detection (exactly 2 unique values
        after cleaning)
```

```

78     binary_categorical_detected = False
79     if dtype == 'category':
80         if num_unique_no_sentinel == 2:
81             binary_categorical_detected = True
82         elif num_unique_no_sentinel == 1 and sentinel_present
83             :
84             binary_categorical_detected = True
85
86     # Multi-class categorical (3+ unique values, object/
87     # category)
88     multi_class_categorical_detected = num_unique_no_sentinel
89     > 2 and dtype == 'category'
90
91     # Single value categorical
92     single_value_detected = num_unique_no_sentinel == 1 and
93     not binary_categorical_detected
94
95     # Special symbols detection (ignore numeric formatting)
96     symbol_detected = clean_vals.astype(str).str.contains(
97         special_symbols_pattern).any()
98
99     # Mixed types detection
100    mixed_detected = len(set(type(v) for v in col_data)) > 1
101    # more than one data type present
102
103    # Determine scenario in order of pipeline
104    if empty_detected:
105        scenario = SCENARIOS[0]
106        handling = "Replace numeric/categorical/empty/blank
107        with -1.0"
108    elif numeric_like and sentinel_detected:
109        scenario = SCENARIOS[1]
110        handling = "Replace numeric/sentinel values with -1.0
111        "
112    elif numeric_like and not sentinel_detected:
113        scenario = SCENARIOS[2]
114        handling = "Keep as numeric"
115    elif censored_detected:
116        scenario = SCENARIOS[3]
117        handling = "Convert numeric ranges to midpoints or
118        keep as categorical"

```

Appendix

```
110     elif binary_categorical_detected:
111         scenario = SCENARIOS[4]
112         handling = "Encode as 0/1 or keep as category"
113     elif multi_class_categorical_detected:
114         scenario = SCENARIOS[5]
115         handling = "One-hot encode, label encode, or keep as
116                     category"
117     elif single_value_detected:
118         scenario = SCENARIOS[6]
119         handling = "Keep as category / constant column"
120     elif symbol_detected:
121         scenario = SCENARIOS[7]
122         handling = "Inspect and preprocess based on context"
123     else:
124         scenario = SCENARIOS[8]
125         handling = "Inspect manually / complex mixed types"
126
127     sample_values = list(col_data.unique()[ :sample_size]) #
128                     sample unique values
129
130     summary.append({
131         'Variable': col,
132         'DataType': dtype,
133         'UniqueValues': num_unique_no_sentinel,
134         'Scenario': scenario,
135         'SuggestedHandling': handling,
136         'SampleValues': sample_values
137     })
138
139     return pd.DataFrame(summary)
```

Appendix D

Second Stage: Deterministic six step cleaning engine

```
1 # Cleaning pipeline to all waves
2 dfs_cleaned = {}
3
4 # flattened dtype df
5 dfs_flat = {wave: flatten_column_types(df) for wave, df in dfs.
6             items()} # flatten dtypes
7
8 # Original df
9 display(scenario_count_table(dfs_flat).style.set_caption("
10     Original Table - pre cleaning")) # display original scenario
11     count table
12
13 # Step 1: Clean empty/missing values
14 dfs_step1 = {wave: clean_empty_missing(df) for wave, df in
15             dfs_flat.items()} # apply step 1 cleaning
16 display(scenario_count_table(dfs_step1).style.hide(axis="index").
17         set_caption("Step 1: Empty / Blank post cleaning"))
18
19 # Step 2: Clean sentinel values
20 dfs_step2 = {wave: clean_sentinel(df) for wave, df in dfs_step1.
21             items()} # apply step 2 cleaning
22 display(scenario_count_table(dfs_step2).style.hide(axis="index").
23         set_caption("Step 2: Sentinel post cleaning"))
24
```

D. SECOND STAGE: DETERMINISTIC SIX STEP CLEANING ENGINE

```
18 # Step 3: Numerical Transformation
19 dfs_step3 = {wave: numerical_transform(df) for wave, df in
    dfs_step2.items()} # apply step 3 cleaning
20 display(scenario_count_table(dfs_step3).style.hide(axis="index").
    set_caption("Step3: Numeric Transformation-post cleaning"))
21
22 # Step 4: Categorical Transformation for censored scenario
23 dfs_step4 = {wave: categorical_transform_S4(df) for wave, df in
    dfs_step3.items()} # apply step 4 cleaning
24 display(scenario_count_table(dfs_step4).style.hide(axis="index").
    set_caption("Step4: Category Transformation_S4-post
    cleaning"))
25
26 # Step 5: Categorical Transformation for binary values
27 dfs_step5 = {wave: categorical_transform_S5(df) for wave, df in
    dfs_step4.items()} # apply step 5 cleaning
28 display(scenario_count_table(dfs_step5).style.hide(axis="index").
    set_caption("Step5: Category Transformation_S5-post
    cleaning"))
29
30 # Step 6: Categorical Transformation for multi class/mixed values
31 dfs_step6 = {wave: categorical_transform_S6(df) for wave, df in
    dfs_step5.items()} # apply step 6 cleaning
32 display(scenario_count_table(dfs_step6).style.hide(axis="index").
    set_caption("Step6: Category Transformation_S6-post
    cleaning"))
33
34 dfs_cleaned = dfs_step6
35
36
37 def show_grouped_vars_for_scenarios(dfs, scenarios_to_show): #
    Function to show grouped variables per scenario across waves
    -> markdown format
38     """
39     Used to inspect the scenario variables/features that need
    handling for transformation/cleaning.
40     A pre and post cleaning helper function to identify variables
    that need specific handling and to
41     Inspect actual changes for a scenario from previous cleaning
    step to current step.
42     """
```

```

43
44 print("#_Category_Transformation_Mapping")
45 print(f"##_Scenario:_{scenarios_to_show}") # Markdown mapping
    dictionary header
46 for wave, df in dfs.items():
47     print(f"\n###_=====_{wave.upper()}_
        =====")
48
49     mapping_df = detect_column_scenario_mapping(df) # Get
        scenario mapping
50
51     for scenario in scenarios_to_show:
52         vars_in_scenario = mapping_df[mapping_df['Scenario']
            == scenario]['Variable'].tolist() # Variables in
            scenario
53         # print(f"\n--- Scenario: {scenario} ---") # Scenario
            header for pipeline git commit
54         print(f"\n---_Total:_{len(vars_in_scenario)}_---") #
            \n
55         if not vars_in_scenario:
56             print("No_variables_found.")
57             continue
58
59         # Dictionary: frozen set(values) -> list of variables
60         groups = {}
61
62         for var in vars_in_scenario:
63             unique_vals = list(dict.fromkeys(df[var].dropna()
                )) # Preserve first-seen order
64             key = tuple(unique_vals) # Use ordered tuple as
                key
65
66             if key not in groups: # Initialise list if key
                not present
67                 groups[key] = []
68             groups[key].append(var) # Append variable to the
                group
69
70         # Print grouped variables in markdown format friendly
            for Notion -> mapping dictionary
71         for value_set, var_list in groups.items():

```

D. SECOND STAGE: DETERMINISTIC SIX STEP CLEANING ENGINE

```
72     print("\n*Variables:*","" + ",".join(var_list)
73           + "'') # \n
74     print(f"*Pre_transformed_values:*_{list(value_set
75           )}')")
76
77     # Pull transformed values from dfs_cleaned /
78     specific step using preserved order
79     try:
80         df_before = df[var_list[0]]
81         df_after = dfs_cleaned[wave][var_list[0]]
82
83         # Build transformed list in same order as
84         original values
85         transformed_vals = []
86         seen = set()
87
88         for orig_val in value_set: # Iterate through
89             original values in order
90             mask = df_before == orig_val
91             if mask.any():
92                 trans_vals = df_after[mask].
93                     unique().tolist() # Get unique
94                     transformed values
95                 if trans_vals:
96                     transformed_vals.append(
97                         trans_vals[0]) # Append
98                         first transformed value
99
100     except Exception:
101         transformed_vals = []
102     print(f"*Transformed_numeric:*_{transformed_vals
103           }'")
104
105     scenarios_to_show = SCENARIOS
106     # scenarios_to_show = ["Clean numeric"]
107     # scenarios_to_show = ["Censored / bracketed numeric"]
108     # scenarios_to_show = ["Binary categorical"]
109     # scenarios_to_show = ["Multi class categorical / mixed"]
110
111     show_grouped_vars_for_scenarios(dfs_cleaned, scenarios_to_show) #
112     prev step == comparison to current step else current step
```

```
transformed/cleaned for ready .md dictionary
```

References

- [1] Kai Petersen, Sairam Vakkalanka, and Ludwik Kuzniarz. Guidelines for conducting systematic mapping studies in software engineering: An update. *Information and Software Technology*, 64:1–18, August 2015. ISSN 0950-5849. doi: 10.1016/j.infsof.2015.03.007. vi, 6, 8
- [2] Barbara Kitchenham, O. Pearl Brereton, David Budgen, Mark Turner, John Bailey, and Stephen Linkman. Systematic literature reviews in software engineering – A systematic literature review. *Information and Software Technology*, 51(1):7–15, January 2009. ISSN 09505849. doi: 10.1016/j.infsof.2008.09.009. vi, 6, 10
- [3] The Irish Longitudinal Study on Ageing. Tilda pseudonymised microdata file release guide v5.5. <https://tilda.tcd.ie/data/documentation/>, 2023. Accessed: 2025-10-09. vi, 28
- [4] James Davis, Silvin P. Knight, Rossella Rizzo, Orna A. Donoghue, Rose Anne Kenny, and Roman Romero-Ortuno. A linear regression-based machine learning pipeline for the discovery of clinically relevant correlates of gait speed reserve from multiple physiological systems. In *2021 29th European Signal Processing Conference (EUSIPCO)*, pages 1266–1270, August 2021. doi: 10.23919/EUSIPCO54536.2021.9616187. vi, 17, 19, 38, 40
- [5] Neal Haddaway, Matthew J Page, Chris C Pritchard, and Luke A McGuinness. PRISMA2020: An R package and Shiny app for producing PRISMA 2020-compliant flow diagrams, with interactivity for optimised digital transparency and Open Synthesis, 2022. vii, 9
- [6] James R. C. Davis, Silvin P. Knight, Orna A. Donoghue, Belinda Hernández, Rossella Rizzo, Rose Anne Kenny, and Roman Romero-Ortuno. Comparison of Gait Speed Reserve, Usual Gait Speed, and Maximum Gait Speed of Adults Aged

REFERENCES

- 50+ in Ireland Using Explainable Machine Learning. *Front Netw Physiol*, 1:754477, 2021. ISSN 2674-0109. doi: 10.3389/fnetp.2021.754477. 1, 10, 11, 17, 19
- [7] Patrick V. Moore, Kathleen Bennett, and Charles Normand. Counting the time lived, the time left or illness? Age, proximity to death, morbidity and prescribing expenditures. *Social Science & Medicine*, 184:1–14, 2017. ISSN 0277-9536. doi: 10.1016/j.socscimed.2017.04.038.
- [8] Ying Huang, Yuhao Su, Ying Jiang, and Meilan Zhu. Sex differences in the associations between blood pressure and anxiety and depression scores in a middle-aged and elderly population: The Irish Longitudinal Study on Ageing (TILDA). *Journal of Affective Disorders*, 274:118–125, September 2020. ISSN 0165-0327. doi: 10.1016/j.jad.2020.05.133. 1
- [9] World Health Organization Europe. New data: Noncommunicable diseases cause 1.8 million avoidable deaths and cost US\$ 514 billion every year, reveals new WHO/Europe report. <https://www.who.int/europe/news/item/27-06-2025-new-data-noncommunicable-diseases-cause-1-8-million-avoidable-deaths-and-cost-us-514-billion-USD-every-year-reveals-new-who-europe-report>, 2025. 1
- [10] Niamh Rice and Charles Normand. The cost associated with disease-related malnutrition in Ireland. *Public Health Nutr*, 15(10):1966–1972, October 2012. ISSN 1475-2727. doi: 10.1017/S1368980011003624. 1
- [11] Caoileann H. Murphy, Eoin Duggan, James Davis, Aisling M. O’Halloran, Silvin P. Knight, Rose Anne Kenny, Sinead N. McCarthy, and Roman Romero-Ortuno. Plasma lutein and zeaxanthin concentrations associated with musculoskeletal health and incident frailty in The Irish Longitudinal Study on Ageing (TILDA). *Experimental Gerontology*, 171:112013, 2023. ISSN 0531-5565. doi: 10.1016/j.exger.2022.112013. 1, 3, 11, 14, 18, 19
- [12] Eamon Laird, Charlotte Lund Rasmussen, Rose Anne Kenny, and Matthew P. Herring. Physical Activity Dose and Depression in a Cohort of Older Adults in The Irish Longitudinal Study on Ageing. *JAMA Netw Open*, 6(7):e2322489, July 2023. ISSN 2574-3805. doi: 10.1001/jamanetworkopen.2023.22489. 1, 14
- [13] C. P. McDowell, R. K. Dishman, M. Hallgren, C. MacDonncha, and M. P. Herring. Associations of physical activity and depression: Results from the Irish Longitudi-

REFERENCES

- nal Study on Ageing. *Experimental Gerontology*, 112:68–75, 2018. ISSN 0531-5565. doi: 10.1016/j.exger.2018.09.004. 1
- [14] Cathal McCrory, Ciaran Finucane, Celia O’Hare, John Frewen, Hugh Nolan, Richard Layte, Patricia M. Kearney, and Rose Anne Kenny. Social Disadvantage and Social Isolation Are Associated With a Higher Resting Heart Rate: Evidence From The Irish Longitudinal Study on Ageing. *J Gerontol B Psychol Sci Soc Sci*, 71(3):463–473, May 2016. ISSN 1079-5014. doi: 10.1093/geronb/gbu163. 1
- [15] Eamon Laird, Matthew P. Herring, Brian P. Carson, Catherine B. Woods, Cathal Walsh, Rose Anne Kenny, and Charlotte Lund Rasmussen. Physical activity for depression among the chronically ill: Results from older diabetics in the Irish longitudinal study on ageing. *Psychiatry Research*, 326:115274, August 2023. ISSN 0165-1781. doi: 10.1016/j.psychres.2023.115274. 3, 11
- [16] Sebastian Moguilner, Silvin P. Knight, James R. C. Davis, Aisling M. O’Halloran, Rose Anne Kenny, and Roman Romero-Ortuno. The Importance of Age in the Prediction of Mortality by a Frailty Index: A Machine Learning Approach in the Irish Longitudinal Study on Ageing. *Geriatrics (Basel)*, 6(3):84, August 2021. ISSN 2308-3417. doi: 10.3390/geriatrics6030084. 3, 10, 39
- [17] Orna A Donoghue, Belinda Hernandez, Matthew D L O’Connell, and Rose Anne Kenny. Using Conditional Inference Forests to Examine Predictive Ability for Future Falls and Syncope in Older Adults: Results from The Irish Longitudinal Study on Ageing. *J Gerontol A Biol Sci Med Sci*, 78(4):673–682, April 2023. ISSN 1758-535X. doi: 10.1093/gerona/glac156. 3, 10, 11
- [18] Richard J. Torraco. Writing Integrative Literature Reviews: Guidelines and Examples. *Human Resource Development Review*, 4(3):356–367, September 2005. ISSN 1534-4843, 1552-6712. doi: 10.1177/1534484305278283. 6
- [19] Tonette S. Rocco and Maria S. Plakhotnik. Literature Reviews, Conceptual Frameworks, and Theoretical Frameworks: Terms, Functions, and Distinctions. *Human Resource Development Review*, 8(1):120–130, March 2009. ISSN 1534-4843, 1552-6712. doi: 10.1177/1534484309332617. 6
- [20] Andrew D Oxman and Gordon H Guyatt. Guidelines for reading literature reviews.

REFERENCES

- [21] Muhammad Azeem Ashraf, Meijia Yang, Yufeng Zhang, Mouna Denden, Ahmed Tlili, Jiayi Liu, Ronghuai Huang, and Daniel Burgos. A Systematic Review of Systematic Reviews on Blended Learning: Trends, Gaps and Future Directions. *Psychology Research and Behavior Management*, 14:1525–1541, October 2021. ISSN null. doi: 10.2147/PRBM.S331741. 6
- [22] Xuezhong Xu, Xue Mingyang, Jie Yang, Hailong Zheng, and Zhifei Che. Tailored machine learning for evaluating the long-term diabetes risk in older individuals: Findings from the Irish Longitudinal Study on Ageing (TILDA). *BMJ Open*, 13(5):e072991, May 2023. ISSN 2044-6055. doi: 10.1136/bmjopen-2023-072991. 10, 11, 16, 17, 18, 19
- [23] Rory Boyle, Lee Jollans, Laura M. Rueda-Delgado, Rossella Rizzo, Görsev G. Yener, Jason P. McMorrow, Silvin P. Knight, Daniel Carey, Ian H. Robertson, Derya D. Emek-Savaş, Yaakov Stern, Rose Anne Kenny, and Robert Whelan. Brain-predicted age difference score is related to specific cognitive functions: A multi-site replication analysis. *Brain Imaging Behav*, 15(1):327–345, February 2021. ISSN 1931-7565. doi: 10.1007/s11682-020-00260-3. 10, 11, 16
- [24] Roy Tzemah-Shahar, Hagit Hochner, Khalil Iktilat, and Maayan Agmon. What can we learn from physical capacity about biological age? A systematic review. *Ageing Research Reviews*, 77:101609, May 2022. ISSN 1568-1637. doi: 10.1016/j.arr.2022.101609. 10
- [25] Morgana A. Shirsath, John D. O’Connor, Rory Boyle, Louise Newman, Silvin P. Knight, Belinda Hernandez, Robert Whelan, James F. Meaney, and Rose Anne Kenny. Slower speed of blood pressure recovery after standing is associated with accelerated brain ageing: Evidence from The Irish Longitudinal Study on Ageing (TILDA). *Cerebral Circulation - Cognition and Behavior*, 6:100212, January 2024. ISSN 2666-2450. doi: 10.1016/j.cccb.2024.100212. 10
- [26] Roman Romero-Ortuno, Peter Hartley, James Davis, Silvin P. Knight, Rossella Rizzo, Belinda Hernández, Rose Anne Kenny, and Aisling M. O’Halloran. Transitions in frailty phenotype states and components over 8 years: Evidence from The Irish Longitudinal Study on Ageing. *Archives of Gerontology and Geriatrics*, 95: 104401, 2021. ISSN 0167-4943. doi: 10.1016/j.archger.2021.104401. 10, 11, 12, 19
- [27] Laura A Bardon, Melanie Streicher, Clare A Corish, Michelle Clarke, Lauren C Power, Rose Anne Kenny, Deirdre M O’Connor, Eamon Laird, Eibhlis M

REFERENCES

- O'Connor, Marjolein Visser, Dorothee Volkert, Eileen R Gibney, and MaNuEL Consortium. Predictors of Incident Malnutrition in Older Irish Adults from the Irish Longitudinal Study on Ageing Cohort—A MaNuEL study. *J Gerontol A Biol Sci Med Sci*, 75(2):249–256, January 2020. ISSN 1079-5006. doi: 10.1093/gerona/gly225. 10, 11, 15, 18
- [28] Zuo Zhang, Lauren Robinson, Robert Whelan, Lee Jollans, Zijian Wang, Frauke Nees, Congying Chu, Marina Bobou, Dongping Du, Ilinca Cristea, Tobias Banaschewski, Gareth J. Barker, Arun L. W. Bokde, Antoine Grigis, Hugh Garavan, Andreas Heinz, Rüdiger Brühl, Jean-Luc Martinot, Marie-Laure Paillère Martinot, Eric Artiges, Dimitri Papadopoulos Orfanos, Luise Poustka, Sarah Hohmann, Sabina Millenet, Juliane H. Fröhner, Michael N. Smolka, Nilakshi Vaidya, Henrik Walter, Jeanne Winterer, M. John Broulidakis, Betteke Maria van Noort, Argyris Stringaris, Jani Penttilä, Yvonne Grimmer, Corinna Insensee, Andreas Becker, Yuning Zhang, Sinead King, Julia Sinclair, Gunter Schumann, Ulrike Schmidt, and Sylvane Desrivières. Machine learning models for diagnosis and risk prediction in eating disorders, depression, and alcohol use disorder. *Journal of Affective Disorders*, 379:889–899, June 2025. ISSN 0165-0327. doi: 10.1016/j.jad.2024.12.053. 10, 11
- [29] Dr Angelina R. Sutin, Martina Luchetti, Damaris Aschwanden, Yannick Stephan, Amanda A. Sesker, and Antonio Terracciano. Sense of meaning and purpose in life and risk of incident dementia: New data and meta-analysis. *Archives of Gerontology and Geriatrics*, 105:104847, 2023. ISSN 0167-4943. doi: 10.1016/j.archger.2022.104847. 10
- [30] doctoralwriting. Creating the literature review: Research questions and arguments, February 2016. 10
- [31] doctoralwriting. Creating the literature review: Research questions and arguments (Part 3 of 4), February 2016. 10
- [32] A. A. Zarudsky and K. I. Prashchayeu. P277: One-legged balance test and falls in old patients with cardiovascular diseases. *European Geriatric Medicine*, 5:S171–S172, 2014. ISSN 1878-7649. doi: 10.1016/S1878-7649(14)70448-6. 11, 16, 17
- [33] Chuanying Huang, Shuqin Sun, Xiaocao Tian, Tong Wang, Tianyu Wang, Haiping Duan, and Yili Wu. Age modify the associations of obesity, physical activity, vision and grip strength with functional mobility in Irish aged 50 and older. *Archives*

REFERENCES

- of Gerontology and Geriatrics*, 84:103895, 2019. ISSN 0167-4943. doi: 10.1016/j.archger.2019.05.020. 11, 12
- [34] Louis Jacob, Karel Kostev, Jae Il Shin, Lee Smith, Hans Oh, Adel S. Abduljabbar, Josep Maria Haro, and Ai Koyanagi. Falls increase the risk for incident anxiety and depressive symptoms among adults aged ≥ 50 years: An analysis of the Irish longitudinal study on ageing. *Archives of Gerontology and Geriatrics*, 114:105098, 2023. ISSN 0167-4943. doi: 10.1016/j.archger.2023.105098. 11, 14
- [35] Kevin McCarthy, Eamon Laird, Aisling M. O’Halloran, Padraic Fallon, Román Romero Ortuño, and Rose Anne Kenny. Association between metabolic syndrome and risk of both prevalent and incident frailty in older adults: Findings from The Irish Longitudinal Study on Ageing (TILDA). *Experimental Gerontology*, 172:112056, 2023. ISSN 0531-5565. doi: 10.1016/j.exger.2022.112056. 12
- [36] Ziggi Ivan Santini, Ai Koyanagi, Stefanos Tyrovolas, Josep Maria Haro, Robert J. Donovan, Line Nielsen, and Vibeke Koushede. The protective properties of Act-Belong-Commit indicators against incident depression, anxiety, and cognitive impairment among older Irish adults: Findings from a prospective community-based study. *Experimental Gerontology*, 91:79–87, 2017. ISSN 0531-5565. doi: 10.1016/j.exger.2017.02.074. 13
- [37] A. L. Juola, M. P. Bjorkman, H. Kautiainen, S. Pylkkanen, U. H. Finne-Soveri, H. Soini, K. H. Pitkälä, and J. S. Bell. O1.17: Nursing staff education to reduce potentially harmful medication use among older people in assisted living facilities: Effects of randomized controlled trial on cognition and falls. *European Geriatric Medicine*, 5:S50–S51, 2014. ISSN 1878-7649. doi: 10.1016/S1878-7649(14)70100-7. 13
- [38] Kevin McCarthy, Eamon Laird, Aisling M. O’Halloran, Cathal Walsh, Martin Healy, Annette L. Fitzpatrick, James B. Walsh, Belinda Hernández, Padraic Fallon, Anne M. Molloy, and Rose Anne Kenny. Association between vitamin D deficiency and the risk of prevalent type 2 diabetes and incident prediabetes: A prospective cohort study using data from The Irish Longitudinal Study on Ageing (TILDA). *EClinicalMedicine*, 53:101654, November 2022. ISSN 2589-5370. doi: 10.1016/j.eclinm.2022.101654. 14
- [39] Kevin McCarthy, Aisling M. O’Halloran, Padraic Fallon, Rose Anne Kenny, and Cathal McCrory. Metabolic syndrome accelerates epigenetic ageing in older adults:

REFERENCES

- Findings from The Irish Longitudinal Study on Ageing (TILDA). *Experimental Gerontology*, 183:112314, 2023. ISSN 0531-5565. doi: 10.1016/j.exger.2023.112314. 15
- [40] Zahra Azizi, Rebecca J. Hirst, Alan O’ Dowd, Cathal McCrory, Rose Anne Kenny, Fiona N. Newell, and Annalisa Setti. Evidence for an association between allostatic load and multisensory integration in middle-aged and older adults. *Archives of Gerontology and Geriatrics*, 116:105155, 2024. ISSN 0167-4943. doi: 10.1016/j.archger.2023.105155. 15
- [41] Soraya Matthews, Mark Ward, Anne Nolan, Charles Normand, Rose Anne Kenny, and Peter May. Predicting mortality in The Irish Longitudinal Study on Ageing (TILDA): Development of a four-year index and comparison with international measures. *BMC Geriatr*, 22:510, June 2022. ISSN 1471-2318. doi: 10.1186/s12877-022-03196-z. 16, 18, 19
- [42] Ziad Obermeyer and Ezekiel J. Emanuel. Predicting the Future - Big Data, Machine Learning, and Clinical Medicine. *The New England Journal of Medicine*, 375(13):1216–1219, September 2016. ISSN 1533-4406. doi: 10.1056/NEJMp1606181. 22